Information Overload Under Rational Learning*

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Abstract

We study how rational agents can perform worse when more information is available. An agent observes a $(d+1)$-dimensional signal vector and chooses a learning rule to estimate an unknown state. The agent trades off a quadratic prediction error with a processing cost that depends on both the rule and the complexity of the information environment. We define *information overload* as an increase in inaccuracy in the optimal mean squared error beyond some finite complexity threshold. Despite having access to unlimited information, we show that if (i) constant (prior-based) rules are costless, and (ii) any sequence of nondegenerate responsive rules becomes too costly as complexity increases, then optimal behavior eventually defaults to a prior-based constant rule, thus establishing necessary and sufficient conditions for overload to occur under rational learning. We also characterize the structure of optimal inference under overload, and apply it to correlation neglect and social learning.

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1 Introduction

Decision-makers increasingly operate in environments characterized by an abundance of information. Improvements in data collection, communication technologies, and social connectivity have increased the number of signals agents can observe and employ towards decision tasks. At the same time, a broad empirical literature documents that decision quality does not necessarily improve monotonically with information availability and may instead worsen when the amount of information becomes overwhelming (Eppler and Mengis, 2004; Roetzel, 2019). This phenomenon, commonly referred to as *information overload*, raises a fundamental question for economic theory: how can rational agents perform worse when they have access to more information?

Canonical economic models of learning and selective information usage do not provide a satisfactory answer to this question. In standard Bayesian decision theory and models of rational inattention, additional information is always beneficial because agents can choose to ignore it at no cost (Blackwell, 1953; Sims, 2003). Similarly, in models of costly information acquisition, while information is costly to acquire, learning accuracy does not degrade with the size of the information environment (Grossman and Stiglitz, 1980; Vives, 1993; Gabaix, 2014). As a result, the literature has yet to identify a mechanism through which information overload arises as an endogenous outcome of optimal learning under rational choice.

Our approach is to model overload as a property of optimal inference under cognitive costs. The main departure from classical learning models is that information processing is subject to costs when signals are aggregated through an endogenously chosen inference rule. We consider an agent who observes a set of $(d+1)$ signals and chooses a learning rule ℓ to estimate a true state. The agent trades off statistical accuracy as measured by the mean squared error against a cognitive cost that can depend both on the complexity of the environment (the number of distinct signal sources $d+1$) and on the responsiveness of the rule itself. This setting distinguishes between two different concepts: the informational content of the data and the endogenous choice of how aggressively the agent extracts and uses that content.

This perspective explains why more information can lower performance, even if agents process it in a fully rational way. While agents can always choose to ignore additional signals, doing so is not the puzzle. The puzzle is why an agent would decide to ignore those signals even though they would lead to increased accuracy. In our framework, each additional signal expands the set of possible learning rules, but also raises the marginal cost of choosing any responsive learning rule. If cognitive costs increase significantly with informational complexity, then beyond a threshold, the agent optimally attenuates responsiveness to signals by assigning lower weights on observed data and shifting that weight toward prior information.

In the limit, inference collapses: optimal behavior converges to a constant rule, and learning accuracy returns to the prior variance.

Our first contribution is to characterize this mechanism at a general level. We provide necessary and sufficient conditions for overload to arise in high-dimensional environments, and illustrate that it takes a clear form. Specifically, as complexity increases, the mean squared error under the optimally chosen learning rule reverts back to the prior benchmark. This outcome does not rely on a specific form of the cost functional or on congestion in attention or communication. It stems from a minimal set of properties: (i) adopting constant rules induces no cost, thus allowing the agent to always use a prior-based rule, and (ii) in highly complex environments, any sequence of learning rules that stays meaningfully responsive as complexity rises, becomes too expensive to implement. Under these conditions, high complexity leads to a complete return to the prior benchmark.

Next, we characterize the structure of optimal learning. We show that when costs are increasing in the variance of learning rules, and the Bayesian posterior mean is affine in signals, the optimal learning rule is also affine in the signals. This result provides a structural microfoundation for the use of linear updating rules, such as DeGroot (1974) learning, in social networks, and demonstrates that these rules are not merely behavioral approximations, but can be understood as exact solutions to the trade-off between accuracy and cognitive burden.

The tractability provided by affine learning allows us to perform a sharp comparative statics in informational complexity. Assuming Gaussian prior information, the optimal affine rule features a simple "cognitive deflator" that scales down the weights to all signals as d grows. As a result, learning performance is generically non-monotone in informational complexity. Specifically, the mean squared error initially falls as additional signals improve statistical accuracy, but eventually increases because expanding the dimension of the information environment raises cognitive load and induces shrinkage toward the prior. This pattern can be summarized as a *U-shaped* mean squared error as a function of information load (equivalently, an inverted-U in accuracy).

Finally, we present two applications of our framework. First, we apply the model to correlation neglect, the empirical regularity that agents treat redundant sources as if they were independent (e.g., Enke and Zimmermann, 2019). In our framework, correlation neglect arises not as an outcome of misspecified learning, but as a consequence of optimal shrinkage in high-complexity environments. Specifically, we show that under the Gaussian-quadratic benchmark, the distance between a *correlation-aware* optimal rule that is fully rational and internalizes correlation structure, and a misspecified rule that neglects correlation, converges to zero as d grows. This follows because as the complexity of the information environment expands, the incremental benefit of choosing weights to reflect the signal covariance structure becomes smaller. Our model therefore predicts that correlation neglect is most likely to be observed in high-complexity settings with many information sources, because the incremental benefit of fully internalizing the covariance structure becomes negligible relative to cognitive burdens. This is in alignment with the experimental findings of Dasaratha and He (2021).

Second, we explore a social learning interpretation of our model by mapping the information dimension d to an agent's degree in a network. Invoking our affine characterization results, our model implies that an agent who observes her own signal and those of her d_i neighbors will adopt an optimal DeGroot-style aggregation rule with a cognitive deflator that uniformly shrinks responsiveness as degree rises. In this network context, this has a clear implication; even holding signal quality fixed, there is a finite optimal degree that equalizes the marginal accuracy gain from additional neighbors to the marginal cognitive cost from processing the additional information. We show that when cognition is sufficiently costly relative to signal quality, additional links are strictly harmful; otherwise, the total loss is minimized at an interior optimal degree; it rises when private signals are noisier and falls when signals are more precise.

This result has two implications. First, if agents choose how many sources to follow based on these cognition-based optimal degrees, heterogeneity in cognitive capacity can generate skewed degree distributions featuring many low-degree agents and a small number of highly connected hubs.

Second, the result has implications for informational interventions. Standard seeding strategies target the most central nodes (Banerjee et al., 2013). In our framework, however, centrality does not automatically make an agent an effective "lever" for additional information. First, high-degree agents place lower marginal weight on any one additional signal (a cognitive deflator), so they may be less responsive at the margin. Second, if central agents are exogenously forced to operate beyond their cognitive optimum ($d_i > d_i^*$), then further information inflows can strictly increase their total loss because processing costs dominate the marginal accuracy gains. Thus, targeting hubs need not maximize behavioral change or diffusion effectiveness, even when additional information would still improve their raw statistical accuracy.

Related Literature

This paper relates to the literature on information processing, learning, and cognitive constraints.

In Bayesian decision theory, more informative signals weakly improve estimates of the true state. In particular, under quadratic losses, Bayes risk does not increase as the information environment expands (Blackwell, 1953). This monotonicity relies on *free disposal of signals*: when an agent receives additional information, she can always ignore it. In other words, the feasible set of decision rules is *nested*: for any $d' > d$ the agent can implement any rule feasible at level d by ignoring the extra coordinates, thus no statistically feasible decision rule is lost when information expands. While our framework preserves this idea, we break monotonicity by allowing the *cost* of implementation to scale with the complexity of the environment. Hence, while the agent retains the option to use any inference rule, doing so becomes more expensive in complex environments.

In rational inattention models, agents choose information structures in an optimal way. Costs are often defined using Shannon mutual information or similar measures of informativeness (Sims, 2003; Matějka and McKay, 2015). A key implication of these models is that agents may rationally ignore some information. However, additional signals do not worsen performance as they can be ignored at zero cost. Similarly, sparsity-based approaches create selective attention by penalizing the use of multiple signals or large attention weights (Gabaix, 2014). These models explain why agents chose to focus only on a small set of signals, but they do not generate information overload. Our approach differs by allowing cognitive costs to scale with the complexity of the information environment itself, so that even optimally sparse rules can become unattractive in sufficiently complex environments.

In models of costly information acquisition, agents choose the quantity or precision of information they purchase, trading off accuracy against acquisition costs. (Grossman and Stiglitz, 1980; Vives, 1993; Ali, 2018). While these models establish rational information avoidance (finite acquisition), posterior uncertainty, and the mean squared error are monotone in informativeness. Therefore, these models do not exhibit a decline in learning accuracy when more information is available; instead, agents merely stop acquiring information. In our framework, the informational environment may expand exogenously and a decline in performance can occur because inference costs increase with complexity, thus leading to an endogenous reversion to prior-based behavior.

Frankel and Kartik (2022) provide a distinct mechanism for why agents may employ less information. They show that when data can be manipulated, decision-making can reduce its informativeness. Thus, a decision-maker can optimally commit to being unresponsive to data in order to improve accuracy. We obtain a similar attenuation result. Information complexity induces lower weights assigned to signals, and agents rely more on prior information as the information environment expands.

A distinct set of models study overload through congestion in attention or communication rather than through the cost of inference. Anderson and De Palma (2012) analyze an attention competition game, showing that an abundance of messages can dilute attention and reduce the effectiveness of information. In De Clippel et al. (2014), consumers can examine only a limited number of markets, so firms compete to be noticed; equilibrium prices endogenously shape attention and introduce an additional layer of cross-market competition. These models consider information overload as an equilibrium outcome resulting from attention competition rather than rational learning. Our framework is complementary; it isolates a mechanism by which learning accuracy itself can deteriorate because inference becomes costly in complex environments, even absent of message congestion.

The social learning literature studies how agents aggregate information from others' actions or beliefs. Models of herding and information cascades show that learning may stop and actions may become uninformative when agents infer only from predecessors' actions (Banerjee, 1992; Bikhchandani et al., 1992). This failure arises from the learning environment itself, not from excessive information. Golub and Jackson (2010) demonstrate how disagreement or inefficient learning can occur on a network under fixed learning rules. Our result also highlight learning inefficiencies but emphasize different aspects. We focus on how having more available signals (for example, through higher degrees) can lower accuracy about the true state as inference costs increase with complexity.

A substantial literature in psychology, marketing, and accounting documents non-monotonic relationships between information quantity and decision performance and commonly reports an inverted U-shaped relationship between the two (Miller, 1956; Chewning Jr and Harrell, 1990). Surveys synthesize this evidence and emphasize that overload is characterized by performance deterioration under high informational loads (Eppler and Mengis, 2004; Roetzel, 2019). This work motivates the paper's central goal: to provide an economic mechanism that delivers such non-monotonic performance under rational choice, without imposing exogenous mistakes or biased information processing. Within our framework, we provide necessary and sufficient conditions for overload to arises under rational learning, and our model replicates the (inverted) U-shape patterns often observed experimentally.

The rest of the paper is organized as follows. Section 2 presents the model. Section 3 characterizes information overload under rational learning. Section 4 derives the structure of optimal learning under information overload. Section 5 conducts comparative statics.

Section 6 applies our results to social learning and correlation neglect. Section 7 concludes.

2 Model

This section presents a learning problem in which an agent faces an information environment of increasing dimensionality. Our modeling approach focuses on two distinct objects: (i) the *statistical environment*, which describes the information available at a given level of complexity, and (ii) the *processing technology*, which captures how costly it is to transform that information into an estimate. This separation helps us to define and analyze information overload without relying on a specific underlying distortion such as limited attention, rational inattention, or misspecification, while still obtaining sharp implications.

Information Environment. There is an unknown state of the world $\theta \in \mathbb{R}$, drawn from a known prior distribution with finite second moment, $E[\theta^2] < \infty$. Let $\mu_\theta := E[\theta]$ and $\sigma_\theta^2 := \mathrm{Var}(\theta)$.

For each integer $d \geq 0$, the agent has access to an information environment consisting of $d+1$ signals,[1]

$$S^{(d)} := (s_0, s_1, \ldots, s_d)^\top \in \mathbb{R}^{d+1}.$$

At this stage, we impose no parametric restrictions on the joint distribution of $(\theta, S^{(d)})$. The signals may be biased, correlated, heterogeneous in quality, or contain redundant components. We assume that all such features are absorbed into the joint law of $(\theta, S^{(d)})$. Throughout, expectations and variances are taken with respect to the joint distribution of $(\theta, S^{(d)})$.

Learning Rule. At any information level d, the agent can select a learning rule

$$\ell : \mathbb{R}^{d+1} \to \mathbb{R},$$

which maps the observed signal vector $S^{(d)}$ into an estimate $\ell(S^{(d)})$ of θ. We restrict attention to rules with finite second moment:

$$\mathcal{L}^{(d)} := \left\{ \ell(S^{(d)}) \mid E[\ell(S^{(d)})^2] < \infty \right\}.$$

This restriction is standard in quadratic-loss environments and ensures that both mean squared error and the variance of the estimate are well-defined.

[1] One may interpret s_0 as an "own" signal and $(s_1, \ldots, s_d)$ as d additional sources. While the analysis does not rely on this interpretation, we explore this possibility in Section 6 where we recast the environment in a social learning context.

Consistency and Nested Environments.

For each $d \geq 0$, the agent observes a $(d+1)$-dimensional signal vector $S^{(d)} = (s_0, \ldots, s_d) \in \mathbb{R}^{d+1}$ about θ. We impose a coherence condition across dimensions: when the environment expands (d increases), the joint distribution of the previously available coordinates does not change.

Formally, for any $d' > d$, let $\pi_{d,d'} : \mathbb{R}^{d'+1} \to \mathbb{R}^{d+1}$ be the coordinate projection $\pi_{d,d'}(s_0, \ldots, s_{d'}) = (s_0, \ldots, s_d)$. *Consistency* means that the joint distribution of $(\theta, S^{(d)})$ coincides with the distribution of $(\theta, \pi_{d,d'}(S^{(d')}))$. Equivalently, under the law of $(\theta, S^{(d')})$, the marginal distribution of $(\theta, s_0, \ldots, s_d)$ is exactly the same as under the law of $(\theta, S^{(d)})$.

This delivers free disposal of additional signals. A learning rule at level d is a measurable mapping $\ell : \mathbb{R}^{d+1} \to \mathbb{R}$, and it induces a report $\ell(S^{(d)})$. Given any such ℓ and any $d' > d$, define its *lift* (extension) to the richer environment by composing with the projection:

$$\ell^{+}(S^{(d')}) \;:=\; \ell\big(\pi_{d,d'}(S^{(d')})\big) \;=\; \ell(s_0, \ldots, s_d).$$

Thus, ℓ^{+} simply ignores the additional coordinates $s_{d+1}, \ldots, s_{d'}$. By consistency, the induced joint distribution of the state and the report is preserved.[2] In particular, any purely statistical performance criterion (e.g., $E[(\ell(S^{(d)}) - \theta)^2]$) can be replicated inside the richer environment by lifting the rule.

However, our objective includes cognitive costs that may depend on the complexity, written $C(\ell; d)$. Therefore, even when ℓ^{+} replicates the same report distribution (and hence the same MSE) at d', it need not replicate the same *total* objective value, because generally $C(\ell^{+}; d') \neq C(\ell; d)$.

Processing Cost. Implementing a learning rule in an environment of size d induces a processing cost

$$C(\ell(S^{(d)}); d) \;\in\; [0, \infty],$$

where allowing for $+\infty$ permits the inclusion of feasibility constraints such as cognitive limits or other restrictions by making some rules too costly to apply.

Decision Problem. The agent chooses a learning rule ℓ to minimize the expected squared

[2]This means, for every Borel set, $B \subseteq \mathbb{R}^2$,

$$\Pr\big((\theta, \ell(S^{(d)})) \in B\big) = \Pr\big((\theta, \ell^{+}(S^{(d')})) \in B\big).$$

.

error in addition to the processing cost of implementing the chosen rule:

$$\min_{\ell(\,S^{(d)}\,)\in\mathcal{L}^{(d)}} \underbrace{E\left[(\ell(S^{(d)})-\theta)^2\right]}_{\text{Inaccuracy}} + \underbrace{C(\ell(S^{(d)});d)}_{\text{Processing cost}}. \tag{P_d}$$

Given d, when an optimal learning rule exists, we denote it as $\ell_d^*(S^{(d)})$. We measure *accuracy* by the mean squared error evaluated at the optimal rule

$$\mathrm{MSE}(d) := E\left[(\ell_d^*(S^{(d)})-\theta)^2\right].$$

Note that $\mathrm{MSE}(d)$ isolates the statistical performance of the chosen rule, while the objective in (P_d) also incorporates processing costs.

3 Information Overload

We impose two assumptions that will allow us to characterize information overload.

Our first assumption delivers a "no-learning" benchmark, which ensures that larger informational environments do not restrict feasible behavior.

Assumption 1 (Costless Constant Rules)**.** *If $\ell(S^{(d)}) \equiv c$ for some constant $c \in \mathbb{R}$, then $C(\ell;d) = 0$ for all d.*

Assumption 1 provides a baseline behavior: the agent can always choose a constant estimate without incurring processing costs.

Since the constant rule $\ell(S^{(d)}) \equiv \mu_\theta$ yields

$$E[(\mu_\theta-\theta)^2] = \mathrm{Var}(\theta) = \sigma_\theta^2 \quad \text{and} \quad C(\mu_\theta;d) = 0,$$

it follows that the minimized objective value in (P_d) is always weakly bounded above by σ_θ^2, and the optimal MSE is uniformly bounded:

$$\mathrm{MSE}(d) \le \sigma_\theta^2 \qquad \text{for all } d \ge 0. \tag{1}$$

This inequality ensures that, no matter the complexity, the agent can always revert back to the prior mean by following a constant rule. This reversion incurs no cost and guarantees a baseline loss that matches the variance of the prior. Moreover, since the prior is already known, mean reversion effectively implies that no learning takes place.

Our second assumption explains how processing costs scale with informational complexity.

Standard arguments suggest that if processing costs did not grow with complexity, more signals would weakly improve statistical accuracy. Information overload, however, requires a force that eventually makes the use of additional information unattractive. The following provides a reduced-form condition for this to occur.

Assumption 2 (Asymptotic Cost Dominance)**.** *For any sequence of learning rules $\{\ell_d\}_{d\geq 0}$ with $\ell_d(S^{(d)}) \in \mathcal{L}^{(d)}$, if*

$$\liminf_{d\to\infty} \operatorname{Var}\big(\ell_d(S^{(d)})\big) \;>\; 0,$$

then

$$\lim_{d\to\infty} C(\ell_d; d) \;=\; +\infty.$$

Assumption 2 states that any sequence of rules that remains *informatively responsive* (i.e., $\liminf_{d\to\infty} \operatorname{Var}(\ell_d(S^{(d)})) > 0$) in high-dimensional environment, becomes prohibitively costly as d grows. Note that this is not a constraint on how many signals the agent can observe. Rather, it is a constraint on the processing cost involved in producing a non-degenerate estimate in high-dimensional environments.

A central distinction in our framework is that cognitive costs scale with the *complexity of the information environment* rather than only with the complexity of the specific rule chosen. A learning rule may be *effectively low-dimensional*—relying on a small subset of cues or a low-dimensional statistic—even when the environment presents a high-dimensional *menu* of potentially relevant signals. If processing costs depended only on the coordinates actively used, then expanding the menu would be innocuous: unused coordinates could be ignored at no additional burden.

In many economic contexts, however, ignoring information is not free. Cues arrive unlabeled and may be redundant or spurious, so implementing even a simple inference procedure requires screening and validation: the agent must determine which cues can be safely discarded and how to guard against double-counting and noise. Our framework captures this *menu burden* in reduced form by allowing costs to increase with the overall dimensionality d of the environment, even when the chosen estimator is effectively simple.[3] Assumption 2 formalizes the implication: as the environment expands, any sequence of rules that remains informatively responsive (i.e., induces nonvanishing variation in the estimate) eventually becomes prohibitively costly.

We define information overload as a global decline in accuracy after reaching a certain finite

[3] This contrasts with sparsity-based bounded rationality models (e.g., Gabaix, 2014) in which costs are tied to the effective dimension (sparsity) of the chosen representation. In such formulations, an agent can typically replicate a sparse rule in a larger environment without incurring additional cost from extra unused coordinates.

complexity threshold.

Definition 1 (Information Overload)**.** *Information overload occurs if there exists a finite* d_1 *such that for every* $d_2 > d_1$*,*

$$\mathrm{MSE}(d_2) \; > \; \mathrm{MSE}(d_1).$$

Our definition of information overload adopts a strong notion. It requires a strict decline in accuracy, not merely diminishing marginal returns. Specifically, overload implies that beyond a certain complexity threshold, the accuracy attainable in richer environments is strictly lower than that attained at moderate complexity. Importantly, the definition does *not* impose a particular shape on the path of $\mathrm{MSE}(d)$: the function may be flat or even oscillatory at intermediate d, so long as sufficiently complex environments eventually deliver strictly lower accuracy than a simpler environment. This interpretation is consistent with conceptualizations of overload in the management and information-systems literature, which emphasize that "too much" information can impair performance by exceeding processing capacity rather than merely failing to provide incremental benefits (Eppler and Mengis, 2004).[4]

To understand when this structural failure occurs, we look at the tension between statistical opportunity and processing costs when complexity is extremely high. Our argument focuses on the agent's performance at the limit. We show that when costs become overwhelming, the rational response to infinite complexity is to disregard signals, which pushes the mean squared error (MSE) back to the no-learning benchmark.

The first step in establishing information overload is to show that any learning rule that improves the MSE relative to the prior benchmark requires an increase in variance.

Lemma 1. *Fix* $\varepsilon > 0$ *and let* ℓ *be any square-integrable random variable. If*

$$E[(\ell - \theta)^2] \; \leq \; \mathrm{Var}(\theta) - \varepsilon,$$

then

$$\mathrm{Var}(\ell) \; \geq \; \frac{\varepsilon^2}{4\,\mathrm{Var}(\theta)}.$$

This result links *statistical improvement* to *responsiveness.* Note that since the rule $\ell \equiv \mu_\theta$ admits an MSE of $\mathrm{Var}(\theta)$, the lemma shows that in order to reduce mean squared error below the one implied by the prior by a nontrivial amount, the estimate must have nontrivial

[4] In Section 5 we study a quadratic-Gaussian benchmark in which $\mathrm{MSE}(d)$ is U-shaped (equivalently, accuracy is single-peaked); see Figure 1. This special case matches the commonly observed inverted-U pattern in empirical discussions of information load and performance.

variance. To improve upon the prior baseline, a rule must react to signals, and the lemma quantifies the minimum variance (or minimum "processing tax") required to achieve such an improvement.

Combining Lemma 1 with Assumptions 1–2 implies that, in sufficiently complex environments, it is optimal to attenuate responsiveness and eventually revert to a prior-based constant rule.

Proposition 1. *Suppose Assumptions 1–2 hold and an optimal learning rule exists for each d. Then,*

$$\lim_{d\to\infty} \mathrm{MSE}(d) \;=\; \mathrm{Var}(\theta).$$

Proposition 1 isolates the collapse component of overload: regardless of the details of the signal structure, sufficiently large informational complexity forces optimal accuracy back to the no-learning benchmark.

Information overload additionally requires that the agent learns *something* at some finite complexity level.

Proposition 2. *Suppose Assumptions 1–2 hold and an optimal learning rule exists for each d. Information overload occurs if and only if both of the following conditions hold:*

(i) *(*Initial informativeness*) There exists a finite d_0 such that $\mathrm{MSE}(d_0) < \mathrm{Var}(\theta)$.*

(ii) *(*Asymptotic collapse*) $\lim_{d\to\infty} \mathrm{MSE}(d) = \mathrm{Var}(\theta)$.*

Proposition 2 makes clear that overload is the combination of a *learning region* at moderate complexity, where accuracy beats the prior benchmark, and an *overload region* at high complexity, where accuracy collapses back to the benchmark.

Several distinct mechanisms can generate this kind of asymptotic collapse. First, in many high-dimensional inference problems, achieving nontrivial accuracy requires estimators whose effective complexity grows with the dimension.[5] Moreover, even when the target parameter is low-dimensional, valid estimation typically requires learning high-dimensional *nuisance* components (auxiliary functions such as conditional expectations, propensity scores, or regression functions) whose complexity can increase with the information environment.[6] If the decision maker faces an implementability or capacity constraint that limits estimator complexity as d increases, then the admissible class of informative learning rules can shrink with d, leaving only low-responsiveness (and in the limit, constant) rules.

[5] See Tibshirani (1996), and Wainwright (2019).

[6] See Vaart (1998) and Newey and McFadden (1994) for semiparametric foundations.

Second, collapse can also happen if agents rely on simplified or misspecified representations of their environment, even though statistically sophisticated rules are feasible. As the true environment grows in complexity, the gap between the truth and the agent's best subjective approximation can widen. In equilibrium, the agent settles on the closest fit to the observed data (e.g., a Berk–Nash equilibrium, as in Esponda and Pouzo (2016)). If the agent's subjective model cannot accommodate high-dimensional correlations, this misspecification can lead the agent to optimally discard the signals and revert to the prior.

Finally, collapse can occur in correctly specified environments with feasible informative rules, if precision demands high sensitivity to the data, and responsiveness becomes more expensive. This is the main idea behind rational inattention and related limited-attention models. In these models, capacity costs on information flow (e.g., mutual information / entropy) lead agents to endogenously choose a coarser information structure and thus attenuate their responsiveness to raw signals (Sims, 2003). In the rest of the paper, we will focus on the third channel and interpret processing costs as *cognitive* costs, which we assume are increasing in the responsiveness of learning rules.

4 Learning Structure

Section 3 established that, under asymptotic cost dominance, accuracy must eventually collapse as informational complexity grows. That argument only relied on the existence of optimal learning rules and not their particular form. To gain further insight, we proceed to characterize the structure of the optimal learning under two standard restrictions. This will in turn allow us to perform comparative statics pertaining to information overload in Section 5.

Throughout, fix an information level $d \geq 0$ and write $S^{(d)} \in \mathbb{R}^{d+1}$ for the signal vector.

4.1 A Convex-Analytic Formulation

To characterize the structure of optimal learning we begin by providing some additional structure on (P_d).

Recall that $\mathcal{L}^{(d)}$ denotes the set of square-integrable estimates that can be generated from $S^{(d)}$. Endow $\mathcal{L}^{(d)}$ with the inner product

$$\langle f, g \rangle := E[f(S^{(d)})g(S^{(d)})],$$

and induced norm $\|f\|_2 := \sqrt{\langle f, f \rangle}$. This makes $(\mathcal{L}^{(d)}, \langle \cdot, \cdot \rangle)$ a Hilbert space.

Then, defining the functional

$$J_d(\ell) := E\big[(\ell(S^{(d)}) - \theta)^2\big] + C(\ell; d), \qquad \ell \in \mathcal{L}^{(d)},$$

allows us to recast (P_d) as functional minimization problem on a Hilbert space.

This formulation is useful for two reasons. First, it clarifies that what the agent is choosing is a measurable mapping from $\mathbb{R}^{d+1}$ to $\mathbb{R}$. Second, it allows us to identify a standard set of conditions under which the minimization problem is well-posed and can be studied using convex analysis.

Assumption 3. *For each $d \geq 0$, the cost functional $C(\cdot; d) : \mathcal{L}^{(d)} \to [0, \infty]$ is:*

(i) Proper: *there exists some $\ell \in \mathcal{L}^{(d)}$ such that $C(\ell; d) < \infty$;*

(ii) Convex: *for all $\ell_1, \ell_2 \in \mathcal{L}^{(d)}$ and $\lambda \in [0, 1]$,*

$$C(\lambda\ell_1 + (1-\lambda)\ell_2; d) \leq \lambda C(\ell_1; d) + (1-\lambda)C(\ell_2; d);$$

(iii) Lower semicontinuous: *if $\ell_n \to \ell$ in $\|\cdot\|_2$, then*

$$C(\ell; d) \leq \liminf_{n\to\infty} C(\ell_n; d).$$

Assumption 3 is common in nonparametric choice problems. Properness rules out the degenerate case where all rules are infeasible. Convexity shows diminishing returns in cognitive effort by stating that implementing the affine mixture of two rules cannot be more expensive than the average of their costs. Lower semicontinuity implies that small perturbations of a learning rule cannot cause a discontinuous drop in cost.

Proposition 3. *Under Assumption 3, for each $d \geq 0$, the problem (P_d) admits a unique minimizer $\ell_d^* \in \mathcal{L}^{(d)}$ (unique up to almost-sure equality).*

Proposition 3 guarantees that the agent's problem is well-defined at each complexity level d, allowing us to talk meaningfully about the path $d \mapsto \ell_d^*$ and $d \mapsto \mathrm{MSE}(d)$.

4.2 Affine Optimality

Let $\mathcal{A}^{(d)}$ denote the set of affine functions of d-dimensional signal vectors:

$$\mathcal{A}^{(d)} := \left\{ a \in \mathcal{L}^{(d)} \mid a(S^{(d)}) = \omega^\top S^{(d)} + c \text{ for some } (\omega, c) \in \mathbb{R}^{d+1} \times \mathbb{R} \right\}.$$

This is a finite-dimensional linear subspace of $\mathcal{L}^{(d)}$ (hence closed).

We now introduce the two restrictions that deliver affine rules as the optimal learning structure.

Assumption 4. *For each $d \geq 0$, the conditional expectation of the state given the signal vector is affine:*

$$E[\theta \mid S^{(d)}] \in \mathcal{A}^{(d)}.$$

Assumption 4 is satisfied in the workhorse benchmark where $(\theta, S^{(d)})$ is jointly Gaussian, and more generally for elliptical families. The assumption states that the agent's *statistically optimal* predictor (the L^2 projection of θ onto the sigma-algebra generated by $S^{(d)}$) is linear in the signals. This rules out environments in which extracting information necessarily requires higher-order nonlinear features of the data. We emphasize that this restriction does *not* assume affine learning. Rather, it states that the best predictor has an affine representation.[7]

Assumption 5. *For each $d \geq 0$, the cost functional is (weakly) increasing in the variance of the induced estimate: for any $f, g \in \mathcal{L}^{(d)}$,*

$$\mathrm{Var}\big(f(S^{(d)})\big) \geq \mathrm{Var}\big(g(S^{(d)})\big) \quad \Longrightarrow \quad C(f; d) \geq C(g; d).$$

Assumption 5 formalizes the idea that, holding the signal dimension d fixed, learning rules that generate greater variability in the agent's report across signal realizations are (weakly) more cognitively costly. Importantly, the assumption does not impose differentiability or any parametric structure. It allows an arbitrary nondecreasing mapping from induced variance into cost, and this mapping may vary with d.

Thus, costs depend on the learning rule only through its induced "responsiveness"[8] as measured by variance (and through d), ruling out additional dependence on other features of ℓ that can vary while keeping variance fixed.

Proposition 4. *Maintain Assumptions 3–5. Then, the unique minimizer ℓ_d^* of (P_d) belongs to $\mathcal{A}^{(d)}$. Equivalently, there exist $(\omega_d^*, c_d^*) \in \mathbb{R}^{d+1} \times \mathbb{R}$ such that*

$$\ell_d^*(S^{(d)}) = (\omega_d^*)^\top S^{(d)} + c_d^*.$$

The result provides a microfoundation for affine learning rules, which are typically assumed as behavioral primitives in learning models (e.g., DeGroot (1974)). Proposition 4 needs to be

[7] When there is no cognitive cost, the best estimator is $\mathbb{E}[\theta \mid S^{(d)}]$ with $S^{(d)}$ signals.

[8] In Section 5, we show a quadratic specification that pins down this ranking sharply.

interpreted as a *dominance* result rather than as a claim that linearity is universally optimal. Any learning rule can be decomposed into an affine component $a(S^{(d)})$ and a residual $r(S^{(d)})$ which is orthogonal to $\mathcal{A}^{(d)}$. The residual cannot improve predictive accuracy about θ beyond what is already captured by the affine conditional expectation, but it necessarily increases dispersion in the realized estimate. Under variance-monotone costs, this makes r weakly harmful on both the dimensions of accuracy and cost, so it is optimally set to zero.

Other than characterizing the form of optimal learning, a key consequence of Proposition 4 is that it reduces the infinite-dimensional problem of choosing an arbitrary measurable learning rule $\ell(\cdot)$, to a finite-dimensional problem over coefficients $(\omega, c) \in \mathbb{R}^{d+1} \times \mathbb{R}$.

Corollary 1. *Maintain Assumptions 3–5. Define the induced cost on coefficients*

$$\widetilde{C}_d(\omega, c) := C\big(\omega^\top S^{(d)} + c;\, d\big).$$

Then (P_d) is equivalent to the convex program

$$\min_{(\omega,c)\in\mathbb{R}^{d+1}\times\mathbb{R}} \; E\big[(\omega^\top S^{(d)} + c - \theta)^2\big] + \widetilde{C}_d(\omega, c),$$

and any optimizer (ω_d^*, c_d^*) yields the unique optimal learning rule $\ell_d^*(S^{(d)}) = (\omega_d^*)^\top S^{(d)} + c_d^*$.

Corollary 1 shows that the choice problem is entirely summarized by the finite-dimensional coefficients (ω, c). This reduction highlights a distinction between the cost of *using* information and the cost of *filtering* the information environment. Standard attention models (e.g., Gabaix (2014)) emphasize effective complexity: they penalize the magnitude or sparsity of the weights ω, capturing how strongly agents respond to particular signals. Our framework captures a closely related class of responsiveness costs—those that are monotone in the induced dispersion of the report, $\mathrm{Var}(\omega^\top S^{(d)})$—so that cognitive effort can be represented by an arbitrary nondecreasing mapping from induced variance into cost.[9]

At the same time, we allow for a distinct *menu burden*. As the information environment expands (indexed by d), the cognitive load of screening and validating potential signals can increase, even if the agent ultimately assigns a nonzero weight to only a few coordinates. Formally, this is why the induced cost $\widetilde{C}_d(\omega, c)$ is permitted to depend explicitly on d in addition to (ω, c). The overload result, therefore, does not come from restricting the agent to complex rules; rather, it captures settings in which identifying a small set of useful cues within a large and noisy menu is itself costly.

[9] Sparsity penalties such as $\sum_i |\omega_i|$ need not be monotone in $\mathrm{Var}(\omega^\top S^{(d)})$ and thus are not literally implied by Assumption 5; we view them as an analogous finite-dimensional formulation that shares the "penalize responsiveness" intuition.

5 Quadratic Cognitive Costs

Sections 2–4 have provided two general results. First, under asymptotic cost dominance, the inaccuracy level must eventually return to the no-learning benchmark as informational complexity grows. Second, under affine conditional expectations and variance-monotone costs, the optimal learning rule is affine.

This section brings together these general ideas into a practical benchmark that allows for closed-form solutions and clear comparative statics. This benchmark has two features: (i) a quadratic variance-based processing cost and (ii) a linear Gaussian signal structure. The results reveal a clear mechanism. As the information environment expands, the agent decreases responsiveness to signals to manage cognitive costs. This creates a U-shaped pattern of inaccuracy after a certain threshold.

Throughout, fix an information level $d \geq 0$ and write $S^{(d)} = (s_0, \ldots, s_d)^\top$.

5.1 Variance-Based Cost and Gaussian Priors

We adopt a quadratic cost that penalizes the dispersion of the estimate across realizations of the signal vector:

$$C_r(\ell(S^{(d)}); d) \ := \ \lambda(d+1)\,\mathrm{Var}\big(\ell(S^{(d)})\big), \tag{2}$$

where $\lambda \geq 0$ controls the extra cost of being responsive. If $\lambda = 0$, the agent does not face any cognitive costs. In this situation, as we demonstrate below, the agent can process the signals in a fully Bayesian way.

The quantity $\mathrm{Var}(\ell(S^{(d)}))$ represents how much the agent's estimate changes in response to data. Constant (prior-based) rules have zero variance, while more responsive rules generate more dispersed estimates. Scaling by $(d+1)$ captures the idea that processing a higher-dimensional input requires additional attention, memory, or computational effort.

Because $C_r(\cdot; d)$ depends on ℓ only through $\mathrm{Var}(\ell(S^{(d)}))$, it is variance-monotone and convex on L^2. Therefore, we can know the cost function satisfies Assumptions 3 and 5.[10] Therefore, the optimal rule under $C_r(\cdot; d)$ is affine. The remainder of this section, therefore, focuses on characterizing the optimal affine coefficients and their comparative statics in d.

With regard to information, we assume that the state and signals satisfy:

[10] We verify that C_r also satisfies the structural assumptions of Section 3 and 4. Assumption 1: If $\ell \equiv c$, then $\mathrm{Var}(\ell) = 0$, so $C_r(\ell; d) = 0$. Assumption 2: if $\mathrm{Var}(\ell) > 0$, then $C_r \to \infty$ as $d \to \infty$. Assumption 3: The functional $\ell \mapsto \mathrm{Var}(\ell) = \|\ell - E[\ell]\|_2^2$ is convex and continuous (hence lower semicontinuous) with respect to the L^2 norm. Assumption 5: Since $\lambda(d+1) \geq 0$, $C_r(\ell; d)$ is nondecreasing in $\mathrm{Var}(\ell)$.

Assumption 6. *The state and signals satisfy:*

$$\theta \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2),$$
$$s_k = \theta + \epsilon_k, \qquad k = 0, 1, \ldots, d,$$
$$(\epsilon_k)_{k=0}^d \ \textit{i.i.d.} \ \ \mathcal{N}(0, \sigma_\epsilon^2), \quad \textit{independent of } \theta.$$

Assumption 6 is imposed only to obtain closed-form expressions. The mechanism identified in Section 4 does not rely on normality; rather, normality lets us write the optimal rule and the corresponding MSE in a transparent way.[11]

5.2 Closed-Form Optimal Rule

Let $\mathbf{1}_{d+1} \in \mathbb{R}^{d+1}$ denote the vector of ones.

Proposition 5. *Under Assumption 6 and the quadratic cost, the unique optimizer of* (P_d) *is affine,*

$$\ell_d^*(S^{(d)}) = (\omega_d^*)^\top S^{(d)} + c_d^*,$$

with symmetric weights $\omega_d^* = \alpha_d \, \mathbf{1}_{d+1}$*, where*

$$\alpha_d = \underbrace{\frac{1}{1 + \lambda(d+1)}}_{\textit{cognitive deflator}} \cdot \underbrace{\frac{\sigma_\theta^2}{\sigma_\varepsilon^2 + \sigma_\theta^2(d+1)}}_{\textit{Bayesian per-signal weight}}, \tag{3}$$

$$c_d^* = \Big(1 - (\omega_d^*)^\top \mathbf{1}_{d+1}\Big)\mu_\theta. \tag{4}$$

Equivalently, let $\bar{s}^{(d)} := \frac{1}{d+1} \sum_{k=0}^d s_k$*, then*

$$\ell_d^*(S^{(d)}) = \mu_\theta + \kappa_d\big(\bar{s}^{(d)} - \mu_\theta\big), \qquad \kappa_d := (d+1)\alpha_d = \frac{1}{1 + \lambda(d+1)} \cdot \frac{\sigma_\theta^2(d+1)}{\sigma_\varepsilon^2 + \sigma_\theta^2(d+1)}. \tag{5}$$

Equation (3) separates two forces. The second factor is the usual Bayesian precision term. If $\lambda = 0$, then each signal would receive the standard Bayesian weight $\sigma_\theta^2/(\sigma_\varepsilon^2 + \sigma_\theta^2(d+1))$. The first factor, $(1 + \lambda(d+1))^{-1}$, is a *cognitive deflator* that shrinks the entire weight vector as either cognitive costs λ or informational dimensionality d rises. Thus, compared to the frictionless Bayesian benchmark, the agent optimally becomes less responsive to any realized signal vector.

We proceed to discuss three implications of Proposition 5.

[11] Assumption 6 implies Assumption 4.

Distribution and Variance. Because ℓ_d^* is an affine function of jointly Gaussian variables, it is also Gaussian. Moreover, the estimator is prior-unbiased ($E[\ell_d^* - \theta] = 0$) and its variance admits a simple form.

Corollary 2. *Maintain Assumption 6 and the quadratic cost specification. Then*

$$\ell_d^*(S^{(d)}) \sim \mathcal{N}\Big(\mu_\theta,\ \mathrm{Var}(\ell_d^*(S^{(d)}))\Big),$$

with

$$\mathrm{Var}(\ell_d^*(S^{(d)})) = \kappa_d^2\Big(\sigma_\theta^2 + \frac{\sigma_\varepsilon^2}{d+1}\Big).$$

Relative to the case where $\lambda = 0$, the variance is uniformly smaller since κ_d is decreasing in λ. This is because cognitive costs induce shrinkage toward the prior mean as captured by a lower κ_d.

Closed-Form MSE. Accuracy is measured by the optimal mean squared error $\mathrm{MSE}(d) = E[(\ell_d^*(S^{(d)}) - \theta)^2]$, which decomposes into a biased-prior component and a noise component.

Corollary 3. *Maintain Assumption 6 and the quadratic cost specification. Then*

$$\mathrm{MSE}(d) = \big(1 - \kappa_d\big)^2\sigma_\theta^2 + \kappa_d^2\frac{\sigma_\varepsilon^2}{d+1}. \tag{6}$$

The first term captures the inaccuracy created by shrinking toward the prior (since $\kappa_d < 1$); the second term captures residual sampling noise from the signal average (σ_ϵ^2).

Asymptotic Collapse. The cognitive deflator forces $\kappa_d \to 0$ as $d \to \infty$ when $\lambda > 0$, producing an explicit instance of the general collapse result from Section 3. Furthermore, as $\lambda \to \infty$, $\kappa_d \to 0$, implying that a high cognitive coefficient induces the same collapse effect.

Corollary 4. *Maintain Assumption 6 and the quadratic cost specification. Then,*

(i) If $\lambda > 0$, then $\kappa_d \to 0$ as $d \to \infty$, and hence

$$\ell_d^*(S^{(d)}) \to \mu_\theta \text{ in } L^2, \qquad \mathrm{MSE}(d) \to \sigma_\theta^2.$$

(ii) If $\lambda = 0$, then $\kappa_d \to 1$ as $d \to \infty$, and $\mathrm{MSE}(d) \to 0$.

Part (i) recovers the no-learning benchmark where large information environments can be self-defeating. Rather than extracting more information, the agent rationally disengages and reverts to the prior mean. Part (ii) shows the fully-Bayesian benchmark where more information always increases accuracy.

5.3 The Burden of Signals and Optimal Complexity

To connect the closed-form solution to the overload narrative, it is useful to separate *accuracy* (or statistical performance) as measured by $\mathrm{MSE}(d)$, from the *total loss* which quantifies the minimized value of (P_d) and which includes cognitive costs. At information level d, denote that latter by

$$L(d) := \min_{\ell \in \mathcal{L}^{(d)}} E[(\ell(S^{(d)}) - \theta)^2] + C_r(\ell; d) = \mathrm{MSE}(d) + \lambda(d+1)\,\mathrm{Var}(\ell_d^*(S^{(d)})).$$

Equations (5) and (6) then imply that an increase in d changes $L(d)$ through two channels: (i) an *information effect* that reduces the noise of the average signal $\bar{s}^{(d)}$ resulting in a lower $\mathrm{MSE}(d)$ and hence a lower $L(d)$, and (ii) a *cognitive load effect* capturing that higher d increases the marginal cost of responsiveness, lowering the optimal κ_d and thereby increasing the bias component of $\mathrm{MSE}(d)$.

The following result provides a full characterization for the behavior of $L(d)$ as we vary the dimension of the information environment.

Proposition 6. *Under Assumption 6 and the quadratic cost, the total loss $L(d)$ changes with d as follows:*[12]

(i) If $\lambda > \frac{\sigma_\varepsilon^2}{2\sigma_\theta^2}$, then $L(d)$ is strictly increasing in d.

(ii) If $\lambda \le \frac{\sigma_\varepsilon^2}{2\sigma_\theta^2}$, then there exists an integer $d^* \ge 0$ such that $L(d)$ is strictly decreasing for $d < d^*$ and strictly increasing for $d > d^*$.

The cutoff between corner and interior complexity is governed by the relative magnitude of cognitive costs and potential accuracy gains. In particular, Proposition 6 implies that the optimal degree collapses to $d^* = 0$ (return to the prior) if and only if

$$\lambda > \frac{\sigma_\varepsilon^2}{2\sigma_\theta^2},$$

that is, when cognition coefficient λ is sufficiently costly relative to the signal-to-prior scale $\sigma_\varepsilon^2/\sigma_\theta^2$. Otherwise, an interior optimum obtains. In that region, the optimal effective number of signals increases with signal noise: noisier signals are individually weaker, but additional independent draws are more valuable because averaging reduces idiosyncratic noise. At the same time, higher-degree agents place lower marginal weight on any one signal, so responsiveness per additional signal declines even as optimal complexity rises.

[12] If $\lambda = 0$, then $L(d)$ is strictly decreasing in d and there is no finite minimizer (equivalently, $d^* = \infty$). Henceforth assume $\lambda > 0$.

Proposition 6 establishes the existence of an optimal complexity level in in terms of the total loss $L(d)$. Information overload, however, is defined purely in terms of *statistical performance*, i.e., in terms of the $\text{MSE}(d)$. We now argue that the same force that eventually makes additional signals unattractive with respect to $L(d)$—the rising marginal cognitive burden that induces stronger prior shrinkage—also drives the long-run deterioration of accuracy.

Formally, Corollary 4 shows that as d approaches infinity, κ_d goes to 0. Consequently, $\text{MSE}(d)$ approaches σ_θ^2. This shows that statistical performance will be back to the no-learning benchmark at high complexity. Therefore, if there is a finite d_0 such that $\text{MSE}(d_0)$ is less than σ_θ^2 (meaning learning is helpful at moderate complexity), then $\text{MSE}(d)$ will eventually start to rise and move back toward σ_θ^2 as d increases. Figure 1 represents this idea and the expected MSE path.

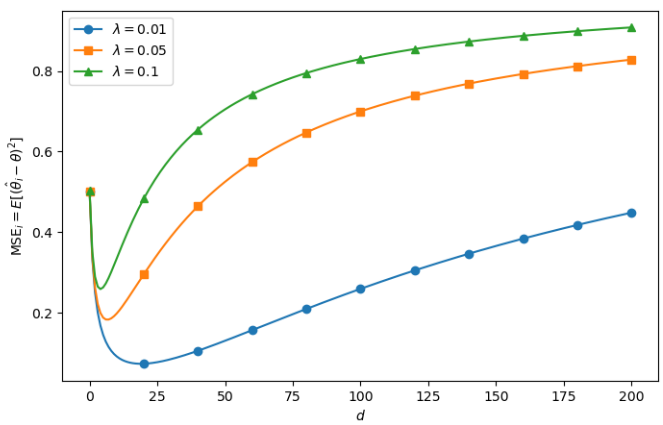

Figure 1: A U-shaped inaccuracy curve (MSE) under quadratic costs. Illustration assumes $\sigma_\theta^2 = \sigma_\varepsilon^2$.

This U-shaped pattern is a clear example of information overload, as defined in Definition 1. Proposition 6 complements this by identifying the optimal operating point for the agent regarding total loss. At the same time, overload is a structural feature of the underlying accuracy curve caused by the same shrinkage mechanism.

Figure 1 shows a mirror image of the familiar "inverted-U" pattern found in the information-overload literature (Eppler and Mengis, 2004; Roetzel, 2019). This literature usually plots *accuracy* (or performance) against informational complexity. It indicates that performance improves at first but eventually declines as cognitive demands increase too much. Our focus is on *inaccuracy*, measured by $\text{MSE}(d)$. Thus, an inverted-U in accuracy is qualitatively equivalent to a U-shaped $\text{MSE}(d)$. The region where more information increases statistical performance relates to a decreasing $\text{MSE}(d)$, while the overload region relates to an increasing MSE(d). Therefore, Figure 1 provides the same qualitative prediction as experimental evidence on information overload, presented through the loss metric suitable for our model.

This behavior of the MSE also provides a theoretical justification for the experimental findings of Dasaratha and He (2021) who show that in a social learning environment, subjects exhibit better accuracy in predicting a state when they are embedded in a sparse network rather than a dense network. The figure aligns with this finding when we interpret d as the degree of an agent embedded in a social network: in a sparse network, subjects have smaller degrees and hence see fewer signals and have smaller cognitive burden as opposed to a dense network. We further discuss this interpretation and its consequences in Section 6.

5.4 Heterogeneous Signal Quality and Selective Weighting

Up to this point, we have assumed that all signals have the same variance. This homoskedastic benchmark implied that the optimal weights across signals are symmetric. However, in many situations, signals may have different reliability levels. We now describe optimal learning when signals have different qualities. This extension shows that the overload mechanism does not require agents to treat all sources the same. Even though cognitive costs reduce the weights to signals overall, the agent still gives more weight to more precise signals.

Assumption 7. *The state satisfies $\theta \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2)$ and the signals satisfy $s_k = \theta + \epsilon_k$ with independent Gaussian noise*

$$\epsilon_k \sim \mathcal{N}(0, \sigma_{\epsilon,k}^2), \qquad k = 0, 1, \ldots, d,$$

independent of θ.

Let $P_d := \sum_{k=0}^{d} \sigma_{\epsilon,k}^{-2}$ denote total precision.

Proposition 7. *Under Assumption 7 and the quadratic cost, the unique optimal rule is affine*

$$\ell_d^*(S^{(d)}) = (\omega_d^*)^\top S^{(d)} + c_d^*, \qquad c_d^* = \big(1 - (\omega_d^*)^\top \mathbf{1}_{d+1}\big)\mu_\theta,$$

with weight components

$$\omega_{d,k}^* = \frac{1}{1+\lambda(d+1)} \cdot \frac{\sigma_\theta^2}{1+\sigma_\theta^2 P_d} \cdot \frac{1}{\sigma_{\epsilon,k}^2}, \qquad k = 0, 1, \ldots, d.$$

Moreover, for any two sources j, k,

$$\sigma_{\epsilon,k}^2 < \sigma_{\epsilon,j}^2 \quad \Longleftrightarrow \quad \omega_{d,k}^* > \omega_{d,j}^*.$$

The optimal weight $\omega^*_{d,k}$ cleanly separates two roles. The common factor $(1 + \lambda(d + 1))^{-1}$ controls *overall responsiveness*: higher cognitive costs shrink all weights simultaneously and move the estimate toward the prior. In contrast, heterogeneity in $\sigma^2_{\varepsilon,k}$ governs *relative allocation*: conditional on choosing to react at all, the agent places higher weight on higher-precision signals, exactly as in standard precision-weighting.

Intuitively, the result reflects the intuition of trust and selective attention in social learning. It shows that a rational player balancing accuracy and cognitive cost will endogenously discount noisy sources and prioritize reliable ones. This is because the weight ω_k is inversely proportional to the noise of signal k, $\sigma^2_{\epsilon,k}$.

It is useful, therefore, to distinguish the roles of two parameters (λ and $\sigma^2_{\epsilon,k}$). The cognitive cost λ controls the agent's overall willingness to learn; a high λ induces information overload, forcing the agent to down-weight all signals regardless of their information quality. In contrast, the signal noise heterogeneity, $\sigma^2_{\epsilon,k}$, considers the relative allocation of attention. Proposition 7 shows that agents remain sophisticated enough to rationally allocate their limited attention by assigning higher weights to reliable signals even if $\lambda > 0$. This distinction is important because λ governs overall responsiveness through a common deflator applied to all weights, while the signal-specific noise variances $\sigma^2_{\epsilon,k}$ determine the relative allocation of responsiveness across signals (via the $1/\sigma^2_{\epsilon,k}$ profile). The total amount of attention used remains endogenous and depends jointly on cognitive costs and signal quality.

6 Applications

We now apply our framework to two relevant settings. First, we use the overload mechanism to provide the micro-foundation for correlation neglect. Second, we provide a network-theoretic interpretation of the information-complexity index d by mapping it to degree in a social network.

6.1 Correlation Neglect

A robust experimental finding is that decision makers often treat redundant signals as if they were independent, failing to discount information that is correlated across sources (e.g., Dasaratha and He, 2021; Enke and Zimmermann, 2019). This phenomenon is known as *correlation neglect* and is commonly interpreted as a behavioral deviation from statistically correct aggregation.

Our framework offers a complementary interpretation that is deliberately *scoped*. We do *not* claim that ignoring correlation is generally optimal inference. Instead, we show that in high-complexity environments—where cognitive costs induce strong shrinkage toward the prior—the behavioral consequences of correctly modeling correlation can become small. In this sense, correlation neglect can emerge as a *nearly behaviorally equivalent* simplification in regimes where agents are optimally pushed toward low responsiveness.

Set Up

We model redundancy by allowing the idiosyncratic noises to be positively correlated across signals. Let

$$\theta \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2), \qquad s_k = \theta + \varepsilon_k, \quad k = 0, 1, \ldots, d,$$

where the noise vector $\varepsilon := (\varepsilon_0, \ldots, \varepsilon_d)^\top$ is independent of θ and distributed as

$$\varepsilon \sim \mathcal{N}\big(0,\ \sigma_\varepsilon^2 \Sigma_\rho\big).$$

We now assume that the correlation matrix Σ_ρ is equicorrelated:

$$\Sigma_\rho := (1-\rho)I + \rho\,\mathbf{1}\mathbf{1}^\top, \qquad \rho \in (0,1).$$

The idiosyncratic noise terms are equicorrelated: $\mathrm{Var}(\varepsilon_k) = \sigma_\varepsilon^2$ and $\mathrm{Corr}(\varepsilon_i, \varepsilon_j) = \rho$ for $i \neq j$. Consequently, the signals $s_k = \theta + \varepsilon_k$ share a common component and correlated noise; larger ρ makes additional signals more redundant (less informative at the margin) about θ.[13]

Let $S^{(d)} = (s_0, \ldots, s_d)^\top$. Under this information structure, the covariance matrix of $S^{(d)}$ is

$$\mathrm{Var}(S^{(d)}) = \sigma_\theta^2\,\mathbf{1}\mathbf{1}^\top + \sigma_\varepsilon^2\big[(1-\rho)I + \rho\,\mathbf{1}\mathbf{1}^\top\big].$$

We maintain the quadratic processing cost used in the benchmark section 5:

$$C_r(\ell; d) = \lambda(d+1)\,\mathrm{Var}(\ell(S^{(d)})), \qquad \lambda \geq 0.$$

By the affine optimality result in Section 4, the optimal rule is affine. Under equicorrelation, symmetry implies that all signals receive the same weight.

Remark 1. *Maintain the Gaussian environment above and the quadratic processing cost. The optimal learning rule takes the form*[14]

$$\ell^*(S^{(d)}) = (\omega^*)^\top S^{(d)} + c^*, \qquad \omega^* = \alpha^* \mathbf{1}, \qquad c^* = \big(1 - (\omega^*)^\top \mathbf{1}\big)\mu_\theta,$$

[13] In particular, $\mathrm{Var}(s_k) = \sigma_\theta^2 + \sigma_\varepsilon^2$ and $\mathrm{Corr}(s_i, s_j) = (\sigma_\theta^2 + \rho\sigma_\varepsilon^2)/(\sigma_\theta^2 + \sigma_\varepsilon^2)$.

[14] We can characterize this learning rule by using the same proof logic in Proposition 5.

where

$$\alpha^* = \frac{\sigma_\theta^2}{(1+\lambda(d+1))\Big(\sigma_\varepsilon^2(1-\rho)+(d+1)(\sigma_\theta^2+\sigma_\varepsilon^2\rho)\Big)}.$$

The expression makes the redundancy channel clear. With d held constant, a higher ρ increases the effective noise in the combined information of the signal vector. The best response is to lower the weight assigned to each signal.

Correlation Neglect as a Misspecified Rule

To model correlation neglect, consider an agent who faces the same cognitive cost but misperceives the covariance structure by treating signals as independent (i.e., sets $\rho = 0$). The agent, therefore, solves the same quadratic-cost problem under a misspecified covariance matrix. The resulting rule is again affine and symmetric.

Definition 2. *The correlation-neglect rule is the affine learning rule obtained by solving the quadratic-cost problem under the misspecified covariance structure $\rho = 0$:*

$$\ell^N(S^{(d)}) = (\omega^N)^\top S^{(d)} + c^N, \qquad \omega^N = \alpha^N \mathbf{1}, \qquad c^N = \big(1-(\omega^N)^\top \mathbf{1}\big)\mu_\theta,$$

with

$$\alpha^N = \frac{\sigma_\theta^2}{(1+\lambda(d+1))\Big(\sigma_\varepsilon^2+(d+1)\sigma_\theta^2\Big)}.$$

The gap between ω^* and ω^N measures how much ignoring correlation distorts the *weights* placed on the signal vector. The central implication below is that its *behavioral distance* from the correlation-aware rule becomes small in high-complexity regimes.

Why the Gap Disappears Under Overload

The key mechanism is the cognitive deflator $(1+\lambda(d+1))^{-1}$: as informational complexity rises, the optimal response to signals is increasingly shrunk toward the prior mean. When d is large (or λ is large), both the correlation-aware and correlation-neglect rules place very little weight on signals, so fine distinctions in covariance modeling have only a small effect on the realized estimate.

Proposition 8. *Let $\rho \in (0,1)$ and $\lambda > 0$. The discrepancy between the correlation-aware* *and correlation-neglect weights satisfies*[15]

$$\|\omega^* - \omega^N\|_2 = O\left(\frac{1}{\lambda(d+1)^{3/2}}\right).$$

Proposition 8 provides a clean comparative static: the behavioral distance between the sophisticated rule (ω^*) and the naive rule (ω^N) shrinks rapidly as informational complexity increases. Therefore, they are nearly behaviorally equivalent. There are two forces behind this convergence.

Large number of signals. When d is large, the additional informational value of correctly accounting for the covariance structure is small relative to the overall shrinkage induced by cognitive costs. Both rules are forced toward low responsiveness, so their weights converge.

High cognitive costs. When λ is large, both rules heavily reduce responsiveness, no matter the details of the signal covariance matrix. In the limit, both rules implement near-constant prior-based estimates, and the distinction between accounting for correlation and ignoring it becomes negligible.

This application gives rise to correlation neglect as a phenomenon induced by informational complexity. In low-complexity situations where λ is small and d is moderate, the optimal learning rule takes into account correlation and information redundancy. The simple approach, however, ignores it. In high-complexity scenarios, however, the behavioral outcomes are similar: both the sophisticated decision maker (ω^*) and the naive one (ω^N) end up in the same low-responsiveness regime, i.e., the no-learning benchmark.

Finally, this mechanism offers a rationalization for why correlation neglect is more prevalent in high-complexity signal environments (e.g., Enke and Zimmermann (2019)): not because agents necessarily misunderstand correlation, but because strong cognitive costs compress behavior toward prior-based estimates, making sophisticated covariance adjustments less consequential.

6.2 Social Learning

This subsection connects the reduced-form information index d to standard objects in network theory by interpreting d as the number of distinct signal sources an agent observes through her social links.

The application serves two purposes. First, it provides a concrete mapping from our earlier comparative statics in d to cross-sectional predictions in networks (via degree heterogeneity

[15] We assume $\lambda > 0$.

and hubs). Second, it connects the model's predictions to important notions in network theory, such as degree differences, the rise of hubs, endogenous influence weights, and DeGroot learning.

Set Up

Let $g = (N, E)$ be an undirected graph with node set $N = \{1, \dots, n\}$. For each agent i, let

$$N_i(g) := \{j \in N : \{i, j\} \in E\} \quad \text{and} \quad d_i := |N_i(g)|$$

denote the neighbor set and (undirected) degree. Agent i observes her own signal and those of her neighbors. Her observed signal vector is[16]

$$S_i(g) := \big(s_j\big)_{j \in N_i(g) \cup \{i\}} \in \mathbb{R}^{d_i+1}.$$

A learning rule for agent i is any measurable map $\ell_i : \mathbb{R}^{d_i+1} \to \mathbb{R}$ with finite second moment under the joint distribution of $(\theta, S_i(g))$.

Given degree d_i and information $S_i(g)$, agent i solves

$$\min_{\ell_i} \; E\big[(\ell_i(S_i(g)) - \theta)^2\big] \; + \; C_r(\ell_i; d_i)\,,$$

where $C_r(\ell_i \cdot; d_i)$ is the same cost functional as in Section 5:

$$C_r(\ell_i; d_i) = \lambda_i (d_i + 1) \; \mathrm{Var}[\ell_i(S_i(g))]\,,$$

allowing the cognitive-cost coefficient $\lambda_i \geq 0$ to vary across agents.

This gives the following correspondence: in earlier sections, d reflected the dimensionality of the information environment. Here, for agent i, the dimensionality equals $d_i + 1$. Thus, the notion of "information level d" can be interpreted as agent i's network degree d_i once we specify that agents observe neighbors' signals.[17]

In the homoskedastic benchmark, symmetry implies that the optimal rule assigns equal

[16] Thus, the dimension of i's information environment is $d_i + 1$.

[17] If agents could also observe their neighbors' neighbors, the relevant information set would expand to

$$S_i^{(2)}(g) := \big(s_i, (s_j)_{j \in N_i(g)}, (s_k)_{k \in \cup_{j \in N_i(g)} N_j(g)}\big),$$

and the cost would be written as $C_r(\ell_i; d_i^{(2)})$ with $d_i^{(2)} := |S_i^{(2)}(g)| - 1$. More generally, d_i can be interpreted as the (known) number of distinct signal sources available to i, which is determined by the underlying network g.

weight to each observed signal. Writing $m_i := d_i + 1$ for the number of observed signals,

$$\ell_i^*(S_i) = \omega_i^* \sum_{j \in N_i(g) \cup \{i\}} s_j + c_i^*,$$

with

$$\omega_i^* = \frac{1}{1+\lambda_i m_i} \cdot \frac{\sigma_\theta^2}{\sigma_\varepsilon^2 + \sigma_\theta^2 m_i}, \qquad c_i^* = \Big(1 - \omega_i^* m_i\Big)\mu_\theta.$$

The first factor is the cognitive deflator; the second is the usual Bayesian precision weight.

Optimal Degrees, Network Structure, and Intervention

The comparative statics in Section 5.3 imply that each agent has an optimal information dimension—and, under the present mapping, an optimal degree—that balances statistical gains from aggregating more signals against the cognitive costs of remaining responsive.

Suppose that each agent can choose how many different sources to observe, for example, by selecting the number of social media accounts to follow or professional contacts to keep. Formally, agent i chooses an integer degree d_i to minimize the total loss resulting from the best learning rule under a signal vector of dimension $d_i + 1$.

Let $L_i(d_i)$ denote the minimized value of the objective given degree d_i, i.e., the total loss when the agent uses the best rule available under a (d_i+1)-dimensional signal vector. Proposition 6 implies two regimes. If cognitive costs are sufficiently high relative to signal quality, then $L_i(d_i)$ is increasing in d_i and the optimal choice is $d_i^* = 0$. Otherwise, $L_i(d_i)$ is single-peaked in accuracy (equivalently U-shaped in loss), yielding an interior minimizer d_i^*.

To focus on the nontrivial case, suppose that optimal degrees are interior for all agents. This is guaranteed when

$$\lambda_i < \frac{\sigma_\varepsilon^2}{2\sigma_\theta^2}, \quad \forall i \in N.$$

Treating $m_i = d_i + 1$ as continuous variable, we get the unique minimizer[18]

$$m_i^* = \sqrt{\frac{\sigma_\varepsilon^2}{\lambda_i \sigma_\theta^2}}, \qquad \text{so} \qquad d_i^* \approx \sqrt{\frac{\sigma_\varepsilon^2}{\lambda_i \sigma_\theta^2}} - 1,$$

with the exact discrete optimum obtained by checking the nearest integers around this value.

The expression shows that d_i^* scales approximately like $\lambda_i^{-1/2}$. Agents with low cognitive costs optimally choose high degrees (they become "information hubs"), whereas agents with high

[18]The optimal degree is calculated in the proof of Proposition 6.

cognitive costs optimally remain in sparse environments to avoid overload. This relationship is illustrated in the figure below, which shows that the optimal degree is a decreasing function of the cognitive cost coefficient when fixing other parameters (σ_ϵ^2 and σ_θ^2).

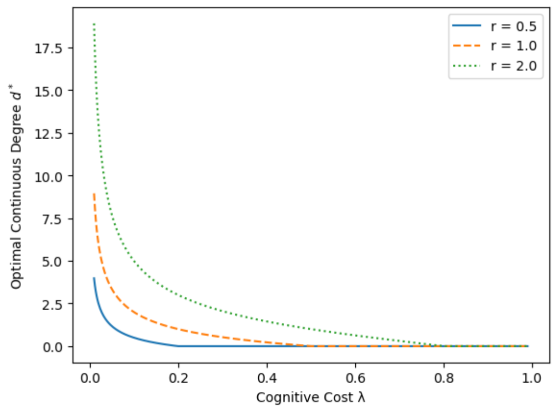

Figure 2: Optimal degree decreases in the cognitive cost λ where $r = \frac{\sigma_\epsilon^2}{\sigma_\theta^2}$.

The figure shows that if we interpret d_i as the number of social links while assuming agents can choose degrees close to d_i^*, then the distribution of cognitive capacity $\{\lambda_i\}_{i \in N}$ induces a degree distribution. If a high proportion of the population is subject to high cognitive costs while the rest face low costs, the mapping $\lambda_i \mapsto d_i^*$ leads to skewed degree distributions. In such a case, we would observe many individuals with low degrees and a small number of hubs with high degrees. This provides a cognition-theoretic explanation for heavy-tailed degree distributions commonly observed in the real-world social networks (Jackson, 2008).

Now, we visit an information intervention environment where the network is exogenous[19]. A standard rationale in diffusion models is to target highly connected influencers because their actions or beliefs have large downstream effects. For example, Banerjee et al. (2013) propose diffusion centrality as a measure of influence in finite-horizon diffusion processes, and show that in early rounds it is closely related to degree. Our framework highlights a distinct intuition about their early rounds centrality: a higher degree is associated with a larger information menu and therefore stronger cognitive shrinkage. As a result, holding signal quality fixed, a high-degree agent places lower marginal weight on any given additional signal, so an extra piece of information delivered to a hub can move that agent's estimate less than the same information delivered to a low-degree agent.

This is a statement about *responsiveness*, not a complete diffusion analysis. In any dynamic model in which downstream propagation is increasing in the initial change in a seeded agent's

[19]Since the network is exogenous, some agents may have a degree over their optimal.

belief or action, our mechanism predicts that "seed the hubs" may be less effective when cognitive costs are significant, because hubs are endogenously less responsive to incremental information. Therefore, diffusion-optimal targeting would require specifying the propagation layer explicitly; here we emphasize the reduced-form implication that degree and influence weights need not align once responsiveness is endogenously attenuated by cognitive costs.

From Unweighted to Weighted Networks

We now extend this social learning perspective by invoking the heterogeneity result of Proposition 7. This allows us to to take into account link weights: while the underlying contact network g stays unweighted and undirected (defining which signals are observable to agents), the optimal learning rule gives continuous, heterogeneous weights to neighbors based on their signal reliability.

Suppose signals differ in quality across neighbors:

$$s_j = \theta + \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}(0, \sigma^2_{\varepsilon,j}), \qquad \varepsilon_j \perp \varepsilon_k \ \ (j \neq k),$$

and maintain the same quadratic cost. Then, Proposition 7 implies that the optimal affine rule assigns a distinct weight to each observed source

$$\ell_i^*(S_i) = \sum_{j \in N_i(g) \cup \{i\}} \omega_{ij}^* s_j + c_i^*, \qquad \omega_{ij}^* \propto \frac{1}{\sigma^2_{\varepsilon,j}},$$

where ω_{ij} is the j-element of a vector ω_i with a common multiplicative shrinkage term driven by (λ_i, d_i).

This expression shows that while the unweighted graph g determines which signals are observable, the endogenous weights $\{\omega_{ij}^*\}$ can be overlaid on g to give a weighted influence network. The entry ω_{ij}^* measures the extent to which neighbor j influences agent i's belief. This construction can thus be understood as an endogenous weighted attention network stemming from rational, cognitive cost-constrained reasoning. Even when overload occurs (high λ_i or large d_i), agents may significantly reduce the weight of all sources while still giving more attention to reliable neighbors.

DeGroot as a Frictionless Limit

Let Ω denote the matrix of influence weights implied by the learning rule, with (i, j) entry $\Omega_{ij} := \omega_{ij}^*$ for $j \in N_i(g) \cup \{i\}$ and $\Omega_{ij} = 0$ otherwise. Under the quadratic cognitive cost, row sums satisfy

$$\sum_{j \in N_i(g) \cup \{i\}} \Omega_{ij} < 1 \quad \text{whenever } \lambda_i > 0,$$

reflecting anchoring on the prior via c_i^* and the cognitive deflator.

In the frictionless limit where cognitive costs disappear and the prior signal is uninformative, the model converges to DeGroot-style averaging:

Corollary 5 (DeGroot limit)**.** *Fix d_i and suppose (i) $\lambda_i \to 0$ and (ii) $\sigma_\theta^2 \to \infty$. Then $c_i^* \to 0$ and*

$$\sum_{j \in N_i(g) \cup \{i\}} \Omega_{ij} \to 1.$$

Moreover, in the homoskedastic benchmark, $\Omega_{ij} \to 1/(d_i + 1)$ for all $j \in N_i(g) \cup \{i\}$.

Therefore, DeGroot-style neighbor averaging is a special case of our framework; it is a conclusion reached when both sources of shrinkage—cognitive variance costs and Bayesian anchoring—are absent.

7 Conclusion

This paper addresses the conflict between rational choice and the real-world observation that having "more information" often reduces decision quality. We build a model where information overload is a byproduct of rational learning under cognitive costs, and show that overload is not the result of irrationality or congestion, but rather a rational outcome from the trade-off between statistical accuracy and cognitive burden.

Our analysis delivers three main insights. First, we provide conditions under which learning "collapses" in high-dimensional environments. When the marginal cost of responsiveness eventually dominates the marginal value of precision, rational agents optimally become unresponsive to data, reverting back to prior-based rules. This mechanism generates the empirically documented inverted-U relationship between information availability and decision accuracy. Second, we characterize the optimal learning strategy. Under variance-monotone costs, the optimal rule is affine, featuring a "cognitive deflator" that uniformly shrinks responsiveness as complexity grows. This provides a microfoundation for linear learning rules often assumed in the social learning literature. Finally, we demonstrate that our framework has sharp implications for network theory and behavioral economics. By connecting the informational complexity environment to the network degree, we show that information overload implies a finite optimal degree, suggesting that highly connected "hubs" may suffer the most from overload. Furthermore, we show that correlation neglect can be rationalized as sophisticated behavior: in complex environments, the loss from ignoring correlation disappears as the agent optimally ignores signals.

These results suggest several avenues for future work. While we analyzed a single decision-maker or a fixed network, our framework naturally extends to equilibrium network formation where agents strategically choose their connectivity to avoid overload. Additionally, our findings have implications for information design. If providing more data can induce a collapse in learning, mechanism designers can consider the construction of optimal information flow structures geared toward enhancing social welfare.

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Appendix: Proofs

Proof of Lemma 1

Expand the mean squared error using the standard bias-variance decomposition:

$$E[(\ell-\theta)^2] \;=\; \mathrm{Var}(\theta)+\mathrm{Var}(\ell)+(E[\ell]-E[\theta])^2-2\,\mathrm{Cov}(\ell,\theta).$$

Since $(E[\ell]-E[\theta])^2 \geq 0$, we can drop the squared bias term. Applying the Cauchy–Schwarz inequality, $\mathrm{Cov}(\ell,\theta) \leq \sqrt{\mathrm{Var}(\ell)\,\mathrm{Var}(\theta)}$, we obtain the lower bound

$$E[(\ell-\theta)^2] \;\geq\; \mathrm{Var}(\theta)+\mathrm{Var}(\ell)-2\sqrt{\mathrm{Var}(\ell)\,\mathrm{Var}(\theta)}.$$

Now, assume the condition $E[(\ell-\theta)^2] \leq \mathrm{Var}(\theta)-\varepsilon$. Substituting this into the inequality above yields

$$\mathrm{Var}(\theta)-\varepsilon \;\geq\; \mathrm{Var}(\theta)+\mathrm{Var}(\ell)-2\sqrt{\mathrm{Var}(\ell)\,\mathrm{Var}(\theta)}.$$

Subtracting $\mathrm{Var}(\theta)$ from both sides and rearranging terms implies

$$2\sqrt{\mathrm{Var}(\ell)\,\mathrm{Var}(\theta)} \;\geq\; \mathrm{Var}(\ell)+\varepsilon \;\geq\; \varepsilon.$$

Squaring both sides and dividing by $4\,\mathrm{Var}(\theta)$ delivers the result:

$$\mathrm{Var}(\ell) \;\geq\; \frac{\varepsilon^2}{4\,\mathrm{Var}(\theta)}.$$

Proof of Proposition 1

By Assumption 1, the constant rule $\ell(S^{(d)}) \equiv \mu_\theta$ is feasible and satisfies $C(\ell; d) = 0$ and $E[(\mu_\theta - \theta)^2] = \mathrm{Var}(\theta)$. Hence the optimal objective value is weakly bounded above by $\mathrm{Var}(\theta)$, and therefore

$$\mathrm{MSE}(d) + C(\ell_d^*; d) \le \mathrm{Var}(\theta) \qquad \text{for all } d \ge 0. \tag{7}$$

Since $C(\ell; d) \ge 0$ for all (ℓ, d),

$$\mathrm{MSE}(d) \;\le\; \mathrm{Var}(\theta) \qquad \text{for all } d \ge 0. \tag{8}$$

We prove $\mathrm{MSE}(d) \to \mathrm{Var}(\theta)$. Fix any $\varepsilon > 0$ and suppose, toward a contradiction, that there exist infinitely many d such that

$$\mathrm{MSE}(d) \le \mathrm{Var}(\theta) - \varepsilon. \tag{9}$$

Let $\{d_k\}_{k\ge1}$ be a strictly increasing sequence of indices satisfying (9). For each k, define the optimal estimator random variable

$$X_{d_k} \;:=\; \ell_{d_k}^*(S^{(d_k)}).$$

Step 1. By Lemma 1, (9) implies a uniform lower bound on dispersion:

$$\mathrm{Var}(X_{d_k}) \ge \underline{v} \quad \text{for all } k, \qquad \text{where } \underline{v} := \frac{\varepsilon^2}{4\,\mathrm{Var}(\theta)} > 0. \tag{10}$$

Step 2. From (7) and $\mathrm{MSE}(d_k) \ge 0$, we obtain

$$C(\ell_{d_k}^*; d_k) \le \mathrm{Var}(\theta) \qquad \text{for all } k. \tag{11}$$

Step 3. For each $d \ge d_1$, let

$$k(d) := \max\{k : \; d_k \le d\}.$$

Let $\pi_{d_{k(d)},d} : \mathbb{R}^{d+1} \to \mathbb{R}^{d_{k(d)}+1}$ denote the coordinate projection $\pi_{d_{k(d)},d}(s_0, \dots, s_d) = (s_0, \dots, s_{d_{k(d)}})$. Define $\hat{\ell}_d \in \mathcal{L}^{(d)}$ by lifting $\ell_{d_{k(d)}}^*$ to environment d and ignoring additional coordinates:

$$\hat{\ell}_d(s_0, \dots, s_d) \;:=\; \ell_{d_{k(d)}}^*\big(\pi_{d_{k(d)},d}(s_0, \dots, s_d)\big) \;=\; \ell_{d_{k(d)}}^*(s_0, \dots, s_{d_{k(d)}}).$$

This rule is feasible by nesting/free disposal.

By the Consistency and Nested Environments environment, the joint distribution of $(\theta, \pi_{d_{k(d)},d}(S^{(d)}))$ coincides with the joint distribution of $(\theta, S^{(d_{k(d)})})$. Consequently, the joint distribution of $(\theta, \hat{\ell}_d(S^{(d)}))$ coincides with that of $(\theta, \ell^*_{d_{k(d)}}(S^{(d_{k(d)})}))$. In particular,

$$\operatorname{Var}(\hat{\ell}_d(S^{(d)})) \;=\; \operatorname{Var}\big(\ell^*_{d_{k(d)}}(S^{(d_{k(d)})})\big) \;\geq\; \underline{v} \qquad \text{for all } d \geq d_1,$$

where the inequality uses (10). Therefore,

$$\liminf_{d\to\infty} \operatorname{Var}(\hat{\ell}_d(S^{(d)})) \geq \underline{v} > 0. \tag{12}$$

Step 4. Along the subsequence $d = d_k$, we have $k(d_k) = k$ and hence $\hat{\ell}_{d_k} = \ell^*_{d_k}$ by construction. Thus, by (11),

$$C(\hat{\ell}_{d_k}; d_k) = C(\ell^*_{d_k}; d_k) \leq \operatorname{Var}(\theta) \qquad \text{for all } k.$$

Hence $C(\hat{\ell}_d; d) \not\to +\infty$ as $d \to \infty$, since it is bounded along the diverging subsequence $\{d_k\}$. Together with (12), this contradicts Assumption 2, which requires that any sequence of rules with $\liminf_{d\to\infty} \operatorname{Var}(\ell_d(S^{(d)})) > 0$ must satisfy $C(\ell_d; d) \to +\infty$.

The contradiction shows that (9) can hold for only finitely many d. Thus for every $\varepsilon > 0$, there exists $D(\varepsilon)$ such that for all $d \geq D(\varepsilon)$,

$$\operatorname{MSE}(d) > \operatorname{Var}(\theta) - \varepsilon,$$

so $\liminf_{d\to\infty} \operatorname{MSE}(d) \geq \operatorname{Var}(\theta)$. Combining with the upper bound (8) yields

$$\lim_{d\to\infty} \operatorname{MSE}(d) = \operatorname{Var}(\theta).$$

■

Proof of Proposition 2

(*Necessity.*) If overload occurs, there exists d_1 such that $\operatorname{MSE}(d) > \operatorname{MSE}(d_1)$ for all $d > d_1$. By (1), $\operatorname{MSE}(d) \leq \operatorname{Var}(\theta)$ for all d. Hence it must be that $\operatorname{MSE}(d_1) < \operatorname{Var}(\theta)$, establishing (i) with $d_0 = d_1$. Condition (ii) follows from Proposition 1 (which requires only Assumptions 1–2).

(*Sufficiency.*) Suppose (i) and (ii) hold. Let d_0 satisfy $\mathrm{MSE}(d_0) < \mathrm{Var}(\theta)$ and define

$$\delta := \frac{\mathrm{Var}(\theta) - \mathrm{MSE}(d_0)}{2} \; > \; 0.$$

By (ii), there exists D such that for all $d \geq D$,

$$\mathrm{MSE}(d) \; > \; \mathrm{Var}(\theta) - \delta \; = \; \frac{\mathrm{Var}(\theta) + \mathrm{MSE}(d_0)}{2} \; > \; \mathrm{MSE}(d_0).$$

Let d_1 be the *largest* minimizer of $\mathrm{MSE}(d)$ on the finite set $\{0, 1, \ldots, D\}$:

$$d_1 \in \arg\min_{0 \leq d \leq D} \mathrm{MSE}(d), \quad \text{chosen so that } d_1 \text{ is maximal among minimizers.}$$

Then for any d with $d_1 < d \leq D$, maximality implies $\mathrm{MSE}(d) > \mathrm{MSE}(d_1)$; and for any $d > D$, the bound above implies $\mathrm{MSE}(d) > \mathrm{MSE}(d_0) \geq \mathrm{MSE}(d_1)$, hence also $\mathrm{MSE}(d) > \mathrm{MSE}(d_1)$. Therefore $\mathrm{MSE}(d) > \mathrm{MSE}(d_1)$ for all $d > d_1$, which is overload.

■

Proof of Proposition 3

Step 1: Properness and lower semicontinuity. By Assumption 3(i), $C(\cdot; d)$ is finite somewhere, and $E[(\ell - \theta)^2] < \infty$ for all $\ell \in \mathcal{L}^{(d)}$ by definition. Hence J_d is proper. The map $\ell \mapsto E[(\ell - \theta)^2]$ is continuous in $\|\cdot\|_2$, and by Assumption 3(iii), $C(\cdot; d)$ is lower semicontinuous, so J_d is lower semicontinuous.

Step 2: Coercivity. Expand

$$E[(\ell - \theta)^2] = E[\ell^2] - 2E[\ell\theta] + E[\theta^2] \geq \|\ell\|_2^2 - 2\|\ell\|_2\|\theta\|_2 + E[\theta^2],$$

where the inequality is Cauchy–Schwarz: $E[\ell\theta] \leq \|\ell\|_2\|\theta\|_2$. Thus $E[(\ell - \theta)^2] \to \infty$ as $\|\ell\|_2 \to \infty$. Since $C(\ell; d) \geq 0$, also $J_d(\ell) \to \infty$ as $\|\ell\|_2 \to \infty$. Hence J_d is coercive.

Step 3: Existence. A proper, coercive, lower semicontinuous convex functional on a Hilbert space attains its minimum on any closed convex domain. Since $\mathcal{L}^{(d)}$ is closed in L^2, J_d attains a minimum on $\mathcal{L}^{(d)}$.

Step 4: Uniqueness. Let $\ell_1 \neq \ell_2$ with $Pro(\ell_1 \neq \ell_2) > 0$ and fix $\lambda \in (0, 1)$. Then

$$E\big[(\lambda\ell_1 + (1 - \lambda)\ell_2 - \theta)^2\big] < \lambda E[(\ell_1 - \theta)^2] + (1 - \lambda)E[(\ell_2 - \theta)^2],$$

because the map $\ell \mapsto E[(\ell-\theta)^2]$ is *strictly* convex on L^2. By convexity of $C(\cdot;d)$,

$$C(\lambda\ell_1 + (1-\lambda)\ell_2; d) \le \lambda C(\ell_1; d) + (1-\lambda)C(\ell_2; d).$$

Adding these two inequalities yields strict convexity of J_d, so J_d cannot have two distinct minimizers. Hence the minimizer is unique up to a.s. equality.

■

Proof of Proposition 4

Fix d and work in L^2 with inner product $\langle X, Y\rangle := E[XY]$. Let $\mathcal{A}^{(d)} := \{\omega^\top S^{(d)} + c : (\omega, c) \in \mathbb{R}^{d+1} \times \mathbb{R}\}$. This is a finite-dimensional (hence closed) linear subspace of $\mathcal{L}^{(d)}$. Let $P_{\mathcal{A}^{(d)}} : \mathcal{L}^{(d)} \to \mathcal{A}^{(d)}$ denote the orthogonal projection.

Take any feasible learning rule $\ell \in \mathcal{L}^{(d)}$ and decompose it as

$$\ell = a + r, \qquad a := P_{\mathcal{A}^{(d)}}(\ell) \in \mathcal{A}^{(d)}, \qquad r := \ell - a \in (\mathcal{A}^{(d)})^\perp.$$

Then $\langle r, h\rangle = 0$ for all $h \in \mathcal{A}^{(d)}$.

Step 1. Let $m := E[\theta \mid S^{(d)}]$ and write $\theta = m + e$ where $E[e \mid S^{(d)}] = 0$. By Assumption 4, we have $m \in \mathcal{A}^{(d)}$. Since $\ell - m$ is measurable with respect to $S^{(d)}$,

$$E[(\ell-m)e] = E\big[E[(\ell-m)e \mid S^{(d)}]\big] = E[(\ell-m)E[e \mid S^{(d)}]] = 0.$$

Therefore,

$$E[(\ell-\theta)^2] = E[(\ell-m)^2] + E[e^2]. \tag{13}$$

Next note that $\ell - m = (a-m) + r$ with $(a-m) \in \mathcal{A}^{(d)}$ and $r \perp \mathcal{A}^{(d)}$, hence

$$E[(\ell-m)^2] = E[(a-m)^2] + E[r^2].$$

Combining this with (13) and observing that $E[(a-\theta)^2] = E[(a-m)^2] + E[e^2]$, we obtain

$$E[(\ell-\theta)^2] = E[(a-\theta)^2] + E[r^2] \;\ge\; E[(a-\theta)^2], \tag{14}$$

with strict inequality whenever $P(r \neq 0) > 0$.

Step 2. Because constants belong to $\mathcal{A}^{(d)}$ (take $\omega = 0$), we have $1 \in \mathcal{A}^{(d)}$. Since $r \perp 1$, it follows that $E[r]=0$. Moreover,

$$\mathrm{Cov}(a,r)=E[(a-E[a])r]=\langle a-E[a],\, r\rangle=0,$$

because $a-E[a]\in\mathcal{A}^{(d)}$ and $r\perp\mathcal{A}^{(d)}$. Hence

$$\mathrm{Var}(\ell)=\mathrm{Var}(a+r)=\mathrm{Var}(a)+\mathrm{Var}(r)+2\,\mathrm{Cov}(a,r)=\mathrm{Var}(a)+\mathrm{Var}(r)\ \geq\ \mathrm{Var}(a). \quad (15)$$

By Assumption 5 and (15),

$$C(\ell;d)\geq C(a;d).$$

Step 3. Combining (14) with cost monotonicity yields

$$J_d(\ell)=E[(\ell-\theta)^2]+C(\ell;d)\ \geq\ E[(a-\theta)^2]+C(a;d)=J_d(a).$$

Thus every feasible ℓ is weakly dominated by its affine projection $a\in\mathcal{A}^{(d)}$.

Finally, let ℓ_d^* be the unique minimizer of (P_d) from Proposition 3. If $\ell_d^*\notin\mathcal{A}^{(d)}$, then its residual r is nonzero with positive probability, so (14) implies $J_d(P_{\mathcal{A}^{(d)}}(\ell_d^*))<J_d(\ell_d^*)$, contradicting optimality. Therefore $\ell_d^*\in\mathcal{A}^{(d)}$, i.e., $\ell_d^*(S^{(d)})=\omega_d^{*\top}S^{(d)}+c_d^*$ for some (ω_d^*,c_d^*).

■

Proof of Proposition 5

Fix $d\geq 0$ and write $m:=d{+}1$. Under Assumption 6, the signal vector satisfies $S^{(d)}=\theta\mathbf{1}_m{+}\varepsilon$, where $\varepsilon\sim\mathcal{N}(0,\sigma_\epsilon^2 I_m)$ is independent of θ. Under the quadratic processing cost,

$$C_r(\ell;d)=\lambda m\,\mathrm{Var}(\ell(S^{(d)})).$$

By Proposition 4 (Affine Optimality), we can restrict attention to affine rules $\ell(S^{(d)})=\omega^\top S^{(d)}+c$.

For any fixed ω, the objective in (P_d) is

$$E\big[(\omega^\top S^{(d)}+c-\theta)^2\big]+\lambda m\,\mathrm{Var}(\omega^\top S^{(d)}+c).$$

The variance term does not depend on c, so the minimizing c is the L^2 projection of $\theta{-}\omega^\top S^{(d)}$ onto constants:

$$0=\frac{\partial}{\partial c}E\big[(\omega^\top S^{(d)}+c-\theta)^2\big]=2\,E\big[\omega^\top S^{(d)}+c-\theta\big],$$

hence

$$c = \mu_\theta - \omega^\top E[S^{(d)}].$$

Since $E[S^{(d)}] = \mu_\theta \mathbf{1}_m$, this yields

$$c = \big(1 - \omega^\top \mathbf{1}_m\big)\mu_\theta.$$

This is exactly the expression in the proposition once we identify $\omega = \omega_d^*$.

Define centered variables $\tilde{\theta} := \theta - \mu_\theta$ and $\tilde{S} := S^{(d)} - \mu_\theta \mathbf{1}_m$. With the optimal c from above, we have

$$\ell(S^{(d)}) - \mu_\theta = \omega^\top \tilde{S}, \qquad \ell(S^{(d)}) - \theta = \omega^\top \tilde{S} - \tilde{\theta}.$$

Let $\Sigma := \mathrm{Var}(\tilde{S}) = \mathrm{Var}(S^{(d)})$. Under Assumption 6,

$$\Sigma = \sigma_\theta^2 \mathbf{1}_m \mathbf{1}_m^\top + \sigma_\epsilon^2 I_m, \qquad \mathrm{Cov}(\tilde{S}, \tilde{\theta}) = \sigma_\theta^2 \mathbf{1}_m.$$

Expanding the objective yields

$$\begin{aligned} E\big[(\omega^\top \tilde{S} - \tilde{\theta})^2\big] &= E[(\omega^\top \tilde{S})^2] - 2E[\omega^\top \tilde{S}\,\tilde{\theta}] + E[\tilde{\theta}^2] \\ &= \omega^\top \Sigma \omega - 2\omega^\top(\sigma_\theta^2 \mathbf{1}_m) + \sigma_\theta^2, \end{aligned}$$

and

$$\mathrm{Var}(\ell(S^{(d)})) = \mathrm{Var}(\omega^\top \tilde{S}) = \omega^\top \Sigma \omega.$$

Therefore the problem reduces to minimizing over $\omega \in \mathbb{R}^m$:

$$\sigma_\theta^2 - 2\sigma_\theta^2\, \omega^\top \mathbf{1}_m + (1 + \lambda m)\, \omega^\top \Sigma \omega.$$

This is a strictly convex quadratic (since $\Sigma \succ 0$ and $1 + \lambda m > 0$), hence the minimizer is unique and satisfies the first-order condition

$$-2\sigma_\theta^2 \mathbf{1}_m + 2(1 + \lambda m)\Sigma\omega = 0, \quad \text{so} \quad \omega = \frac{\sigma_\theta^2}{1 + \lambda m}\, \Sigma^{-1} \mathbf{1}_m.$$

Write $\Sigma = \sigma_\epsilon^2 I_m + \sigma_\theta^2 \mathbf{1}_m \mathbf{1}_m^\top$. By the Sherman–Morrison formula,

$$(\sigma_\epsilon^2 I_m + \sigma_\theta^2 \mathbf{1}\mathbf{1}^\top)^{-1} = \frac{1}{\sigma_\epsilon^2} I_m - \frac{\sigma_\theta^2}{\sigma_\epsilon^2\big(\sigma_\epsilon^2 + \sigma_\theta^2 m\big)}\, \mathbf{1}\mathbf{1}^\top,$$

hence

$$\Sigma^{-1}\mathbf{1}_m = \left(\frac{1}{\sigma_\epsilon^2} - \frac{\sigma_\theta^2 m}{\sigma_\epsilon^2(\sigma_\epsilon^2+\sigma_\theta^2 m)}\right)\mathbf{1}_m = \frac{1}{\sigma_\epsilon^2+\sigma_\theta^2 m}\,\mathbf{1}_m.$$

Substituting into $\omega = \frac{\sigma_\theta^2}{1+\lambda m}\Sigma^{-1}\mathbf{1}_m$ gives

$$\omega_d^* = \alpha_d\,\mathbf{1}_m, \qquad \alpha_d = \frac{1}{1+\lambda(d+1)}\cdot\frac{\sigma_\theta^2}{\sigma_\epsilon^2+\sigma_\theta^2(d+1)}.$$

Finally, using the optimal c yields

$$c_d^* = \big(1-(\omega_d^*)^\top\mathbf{1}_m\big)\mu_\theta.$$

■

Proof of Proposition 6

Maintain Assumption 6 and the quadratic cost. For each $d \geq 0$, define the minimized *total loss*

$$L(d) := \min_{\ell\in\mathcal{L}^{(d)}} \left\{E\big[(\ell(S^{(d)})-\theta)^2\big] + \lambda(d+1)\operatorname{Var}(\ell(S^{(d)}))\right\}.$$

By Proposition 5, the minimizer is affine with coefficient vector ω_d^*, and the reduced problem in ω is a strictly convex quadratic. A standard quadratic-completion argument yields the minimized value

$$L(d) = \sigma_\theta^2 - \frac{\sigma_\theta^4}{1+\lambda(d+1)}\,\mathbf{1}_m^\top\Sigma^{-1}\mathbf{1}_m, \qquad m = d+1, \qquad \Sigma = \sigma_\epsilon^2 I_m + \sigma_\theta^2\mathbf{1}_m\mathbf{1}_m^\top.$$

From the computation in the proof of Proposition 5, $\Sigma^{-1}\mathbf{1}_m = \frac{1}{\sigma_\epsilon^2+\sigma_\theta^2 m}\mathbf{1}_m$, hence

$$\mathbf{1}_m^\top\Sigma^{-1}\mathbf{1}_m = \frac{m}{\sigma_\epsilon^2+\sigma_\theta^2 m}.$$

Therefore,

$$L(d) = \sigma_\theta^2 - \frac{\sigma_\theta^4(d+1)}{(1+\lambda(d+1))\big(\sigma_\epsilon^2+\sigma_\theta^2(d+1)\big)}.$$

Let $m = d+1 > 0$ be real valued and define

$$\widetilde{L}(m) := \sigma_\theta^2 - \frac{\sigma_\theta^4 m}{(1+\lambda m)(\sigma_\epsilon^2+\sigma_\theta^2 m)}.$$

Differentiating yields

$$\widetilde{L}'(m) = -\frac{\sigma_\theta^4\big(\sigma_\epsilon^2 - \lambda\sigma_\theta^2 m^2\big)}{(\sigma_\epsilon^2 + \sigma_\theta^2 m)^2(1+\lambda m)^2},$$

so $\widetilde{L}'(m) < 0$ when $m < \sqrt{\sigma_\epsilon^2/(\lambda\sigma_\theta^2)}$ and $\widetilde{L}'(m) > 0$ when $m > \sqrt{\sigma_\epsilon^2/(\lambda\sigma_\theta^2)}$ (for $\lambda > 0$). Hence $\widetilde{L}$ is strictly decreasing up to the unique minimizer

$$m^* := \sqrt{\frac{\sigma_\epsilon^2}{\lambda\sigma_\theta^2}},$$

and strictly increasing thereafter. (When $\lambda = 0$, $\widetilde{L}$ is strictly decreasing in m, implying it admits a corner solution.)

Because $\widetilde{L}$ is decreasing then increasing, the integer sequence $L(d)$ is unimodal: it strictly decreases while $d+1 < m^*$ and strictly increases once $d+1 > m^*$. Thus the only remaining question is whether the sequence is already increasing at $d = 0$.

Compute the first increment:

$$L(1) - L(0) = \widetilde{L}(2) - \widetilde{L}(1).$$

Using the closed form above, we have

$$\widetilde{L}(2) - \widetilde{L}(1) = -\frac{\sigma_\theta^4\big(\sigma_\epsilon^2 - 2\lambda\sigma_\theta^2\big)}{(1+2\lambda)(\sigma_\epsilon^2 + 2\sigma_\theta^2)(1+\lambda)(\sigma_\epsilon^2 + \sigma_\theta^2)}.$$

The denominator is strictly positive, so the sign is determined by $\sigma_\epsilon^2 - 2\lambda\sigma_\theta^2$.

Case (i): $\lambda > \sigma_\epsilon^2/(2\sigma_\theta^2)$**.** Then $\widetilde{L}(2) > \widetilde{L}(1)$, i.e. $L(1) > L(0)$. Moreover, $\lambda > \sigma_\epsilon^2/(2\sigma_\theta^2)$ implies $m^* < \sqrt{2} < 2$. Since $\widetilde{L}$ is strictly increasing for all $m \geq 2$, it follows that $L(d)$ is strictly increasing for every integer $d \geq 0$.

Case (ii): $\lambda \leq \sigma_\epsilon^2/(2\sigma_\theta^2)$**.** Then $\widetilde{L}(2) \leq \widetilde{L}(1)$, i.e. $L(1) \leq L(0)$, so the sequence does not start in the increasing region. Because $\widetilde{L}$ is unimodal and tends to σ_θ^2 as $m \to \infty$, there exists at least one minimizer over integers; unimodality implies that there is an integer $d^* \geq 0$ such that $L(d)$ strictly decreases for $d < d^*$ and strictly increases for $d > d^*$ (with possible tie only in the knife-edge equality case).

■

Proof of Proposition 7

By Proposition 4, the optimal rule is affine: $\ell(S^{(d)}) = \omega^\top S^{(d)} + c$. Indeed, under Assumption 7, the vector $(\theta, S^{(d)})$ is jointly Gaussian, so the conditional expectation is affine, and with quadratic cognitive costs the induced cost is variance-monotone and satisfies Assumption 3.

As in the proof of Proposition 5, the optimal intercept satisfies $c = \mu_\theta - \omega^\top E[S^{(d)}]$ and therefore $c = \big(1 - \omega^\top \mathbf{1}_m\big)\mu_\theta$. Let $\tilde{S} = S^{(d)} - \mu_\theta \mathbf{1}_m$ and $\tilde{\theta} = \theta - \mu_\theta$. Then the objective becomes

$$\sigma_\theta^2 - 2\sigma_\theta^2\, \omega^\top \mathbf{1}_m + (1 + \lambda m)\, \omega^\top \Sigma\, \omega,$$

where now

$$\Sigma = \mathrm{Var}(S^{(d)}) = \sigma_\theta^2 \mathbf{1}_m \mathbf{1}_m^\top + D, \qquad D := diag(\sigma_{\epsilon,0}^2, \ldots, \sigma_{\epsilon,d}^2).$$

Strict convexity implies the unique minimizer satisfies

$$(1 + \lambda m)\Sigma\omega = \sigma_\theta^2 \mathbf{1}_m, \qquad \text{so} \qquad \omega = \frac{\sigma_\theta^2}{1 + \lambda m}\, \Sigma^{-1} \mathbf{1}_m.$$

To compute $\Sigma^{-1}\mathbf{1}_m$, apply the Woodbury identity with $\Sigma = D + \sigma_\theta^2 \mathbf{1}\mathbf{1}^\top$:

$$\Sigma^{-1} = D^{-1} - D^{-1}\mathbf{1} \left(\frac{1}{\sigma_\theta^2} + \mathbf{1}^\top D^{-1} \mathbf{1} \right)^{-1} \mathbf{1}^\top D^{-1}.$$

Let $P_d := \mathbf{1}^\top D^{-1} \mathbf{1} = \sum_{k=0}^{d} \sigma_{\epsilon,k}^{-2}$. Then

$$\begin{aligned} \Sigma^{-1}\mathbf{1} &= D^{-1}\mathbf{1} - D^{-1}\mathbf{1} \left(\frac{1}{\sigma_\theta^2} + P_d \right)^{-1} \mathbf{1}^\top D^{-1} \mathbf{1} \\ &= D^{-1}\mathbf{1} \left(1 - \frac{P_d}{\frac{1}{\sigma_\theta^2} + P_d} \right) = \frac{1}{1 + \sigma_\theta^2 P_d}\, D^{-1}\mathbf{1}. \end{aligned}$$

Thus

$$\omega = \frac{\sigma_\theta^2}{1 + \lambda m} \cdot \frac{1}{1 + \sigma_\theta^2 P_d}\, D^{-1}\mathbf{1},$$

so each component satisfies

$$\omega_{d,k}^* = \frac{1}{1 + \lambda(d+1)} \cdot \frac{\sigma_\theta^2}{1 + \sigma_\theta^2 P_d} \cdot \frac{1}{\sigma_{\epsilon,k}^2}, \qquad k = 0, 1, \ldots, d.$$

The monotonicity statement $\sigma_{\epsilon,k}^2 < \sigma_{\epsilon,j}^2 \Leftrightarrow \omega_{d,k}^* > \omega_{d,j}^*$ is immediate from the last display.

■

Proof of Proposition 8

Fix $\rho \in (0,1)$. Let $m = d+1$. By Remark 1, the optimal (correlation-aware) coefficient vector is $\omega^* = \alpha^* \mathbf{1}_m$ with

$$\alpha^* = \frac{\sigma_\theta^2}{(1+\lambda m)\big(\sigma_\epsilon^2(1-\rho) + m(\sigma_\theta^2 + \sigma_\epsilon^2\rho)\big)}.$$

The correlation-neglect rule treats the signals as independent (i.e. uses $\rho = 0$ in the same optimization), so it has $\omega^N = \alpha^N \mathbf{1}_m$ with

$$\alpha^N = \frac{\sigma_\theta^2}{(1+\lambda m)\big(\sigma_\epsilon^2 + m\sigma_\theta^2\big)}.$$

Therefore,

$$\|\omega^* - \omega^N\|_2 = \sqrt{m}\,|\alpha^* - \alpha^N|.$$

Compute the difference:

$$\begin{aligned}
\alpha^* - \alpha^N &= \frac{\sigma_\theta^2}{1+\lambda m}\left(\frac{1}{\sigma_\epsilon^2(1-\rho) + m(\sigma_\theta^2 + \sigma_\epsilon^2\rho)} - \frac{1}{\sigma_\epsilon^2 + m\sigma_\theta^2}\right) \\
&= \frac{\sigma_\theta^2}{1+\lambda m} \cdot \frac{\big(\sigma_\epsilon^2 + m\sigma_\theta^2\big) - \big(\sigma_\epsilon^2(1-\rho) + m(\sigma_\theta^2 + \sigma_\epsilon^2\rho)\big)}{\big(\sigma_\epsilon^2(1-\rho) + m(\sigma_\theta^2 + \sigma_\epsilon^2\rho)\big)\big(\sigma_\epsilon^2 + m\sigma_\theta^2\big)} \\
&= \frac{\sigma_\theta^2}{1+\lambda m} \cdot \frac{\sigma_\epsilon^2\rho(1-m)}{\big(\sigma_\epsilon^2(1-\rho) + m(\sigma_\theta^2 + \sigma_\epsilon^2\rho)\big)\big(\sigma_\epsilon^2 + m\sigma_\theta^2\big)}.
\end{aligned}$$

Taking absolute values and using $\rho \in [0,1)$,

$$|\alpha^* - \alpha^N| = \frac{\sigma_\theta^2\sigma_\epsilon^2\rho(m-1)}{(1+\lambda m)\big(\sigma_\epsilon^2(1-\rho) + m(\sigma_\theta^2 + \sigma_\epsilon^2\rho)\big)\big(\sigma_\epsilon^2 + m\sigma_\theta^2\big)}.$$

Since $\sigma_\epsilon^2 + m\sigma_\theta^2 \geq m\sigma_\theta^2$ and $\sigma_\epsilon^2(1-\rho) + m(\sigma_\theta^2 + \sigma_\epsilon^2\rho) \geq m\sigma_\theta^2$ for $\rho \geq 0$, we obtain the bound

$$|\alpha^* - \alpha^N| \leq \frac{\sigma_\theta^2\sigma_\epsilon^2\rho(m-1)}{(1+\lambda m)\,(m\sigma_\theta^2)\,(m\sigma_\theta^2)} = \frac{\sigma_\epsilon^2\rho}{1+\lambda m} \cdot \frac{m-1}{m^2} \cdot \frac{1}{\sigma_\theta^2}.$$

Multiplying by $\sqrt{m}$ gives

$$\|\omega^* - \omega^N\|_2 \leq \frac{\sigma_\epsilon^2\rho}{\sigma_\theta^2} \cdot \frac{1}{1+\lambda m} \cdot \frac{\sqrt{m}(m-1)}{m^2} = O\!\left(\frac{1}{(1+\lambda(d+1))(d+1)^{1/2}}\right).$$

■